VIETNAMESE LEGAL ONTOLOGY AND KNOWLEDGE GRAPH

VLOnto playbook

Build, validate, query and use the Vietnamese legal knowledge graph

Related paper: *Vietnamese Legal Ontology: Every legal answer, traced to the article that grounds it* (Bui and Tran Minh, 2026), arXiv:2611.11426

Release	v1 (version 1.0.0)
Build date	2026-11-12
Author	tmquan

Key figures of VLOnto v1 (1.0.0, this release), knowledge-graph build of 2026-11-12

40,816 legal documents	**436,309** articles (Điều)	**15,035** dated change events
109,543 legal Q&As (metadata)	**6,963,753** RDF triples	**20 / 20** build checks passed

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Not legal advice and not an official legal text. Relations are extracted automatically and may be wrong; read the source documents.

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1. Scope, audience and disclaimer

VLOnto is an OWL ontology of Vietnamese law and a knowledge graph built with it from four public datasets of Vietnamese legal material. In VLOnto v1 (version 1.0.0, this release), the graph connects 40,816 legal documents and their 436,309 articles (*Điều*) with the 15,035 dated events that replaced, amended or repealed them, with 99,038 legal-basis links and 268,563 citation links between documents, and with 109,543 published legal Q&As, 16,701 legal terms and 681 legal news items. The build writes 21 parquet tables, a full RDF dump of 6,963,753 triples and a browser-sized RDF subset of 49,846 triples. VLOnto is published as code and a website: the GitHub repository holds the build code, the ontology with SHACL shapes and documentation, a website with 3D playgrounds and a SPARQL playground, and this playbook. The data stays in the four source datasets, and the knowledge graph is built from them on your own machine (section 2.2).

Court decisions join in v2. VLOnto v1 is built from the four thuvienphapluat.vn datasets only. The court datasets `anle-toaan-gov-vn` and `cbba-toaan-gov-vn` (court decisions and precedents, with the provisions they apply) are planned for VLOnto v2 (version 2.0.0); the judiciary classes and properties stay in the ontology with a scope note that says so (section 4.2). *Bản án, quyết định của Tòa án và án lệ sẽ được bổ sung trong VLOnto v2.*

Who this is for. Engineers who build legal search, question answering or agents over Vietnamese law and need structured validity, citation and Q&A-usage data; legal-technology practitioners who want to query that data without building it; and maintainers who rebuild and release VLOnto. The playbook assumes working knowledge of Python and SQL; SPARQL and OWL are introduced where they are used.

What it covers. How to open the offline website and query the files (section 2), how the data flows from the source datasets to the outputs of the build (section 3), the ontology (section 4), how to rebuild the graph (section 5) and how it was validated (section 6), query recipes with their real results (section 7), how to use VLOnto inside the Vietnamese legal assistant agent (section 8), the website (section 9), reproducibility and maintenance (sections 10 and 11) and the known limits (section 12). How to cite VLOnto is in section 1.1, and the references follow section 12. The appendices are generated from the ontology file, the data dictionary and the licence policy.

Where the numbers come from. Every number in this playbook is read from the build reports of 2026-11-12 by `playbook/tools/gather_facts.py`, which also runs every query recipe and code example against that build and stores their output; `playbook/tools/render_md.py` then fills this text. No number is typed by hand, so a rebuild of the graph followed by a rebuild of the playbook updates all of them. All of them describe VLOnto v1 (1.0.0), this release.

> **Not legal advice.** VLOnto is a research and engineering resource. It is not an official legal text and not legal advice. Its relations are extracted automatically from public sources and can be wrong; the validity of 28,660 of the 40,816 documents (70.2%) is unknown. Read the source document (every document, Q&A and term keeps its source URL) and ask a qualified Vietnamese lawyer before relying on any legal conclusion.

1.1 How to cite

The approach is described in the paper *Vietnamese Legal Ontology: Every legal answer, traced to the article that grounds it* (Bui and Tran Minh, 2026, arXiv:2611.11426; see References). Numbers in this playbook are computed from the VLOnto v1 (1.0.0) release and can differ from the paper's. Cite the paper with this BibTeX entry; the repository also has a `CITATION.cff` file:

```
@misc{bui2026vietnamese,
  title         = {Vietnamese Legal Ontology: Every legal answer, traced to the article
                   that grounds it},
  author        = {Bui, Khac-Hoai Nam and Tran Minh, Quan},
  year          = {2026},
  eprint        = {2611.11426},
  archivePrefix = {arXiv},
  primaryClass  = {cs.CL},
```

```
  url           = {https://arxiv.org/abs/2611.11426}
}
```

2. Quick start

2.1 Open the offline playground

The website is published at https://tmquan.github.io/VLOnto/. It also runs from a folder on disk, without a server and without network access:

1. Clone the repository https://github.com/tmquan/VLOnto or download it as a ZIP file.
2. Double-click `site/index.html` in the repository folder: it opens the 3D ontology playground as a `file://` URL. The overview of VLOnto, with the key counts of the build, is `site/about.html`.
3. The other pages are the legal-network playground, the SPARQL playground, the dashboard, the question traces, the ontology documentation, this playbook and the downloads page; section 9 lists every page with its file name.

All scripts, fonts and data are inside the folder; the pages load no external resources. The site is checked with headless Chrome over `file://` and `http://` (`tools/site_check.py`); Safari and Firefox are not machine-tested, so open the site in them before relying on them (section 12).

2.2 Get the code and the data

VLOnto is published as code and a website, not as a data release. The GitHub repository https://github.com/tmquan/VLOnto holds the build code (`vlonto/`), the corpus pipeline (`pipeline/`), the ontology (`ontology/`), this playbook (`playbook/`) and the website (`site/`), which GitHub Pages serves at https://tmquan.github.io/VLOnto/. The data comes from the four source datasets on Hugging Face (section 3): `python -m vlonto.cli all` builds the knowledge graph from them on your machine (section 5) and writes the parquet tables to `build/kg/` and the RDF dumps to `build/rdf/`. The build reads the processed corpus of the vn-legal project (section 5.1), which `pipeline/` produces from the source datasets (see `pipeline/README.md`). The ontology and the browser subset need no build: the website's downloads page offers them.

File	Size	Where to get it
`kg/*.parquet` (21 tables)	27.9 MB	local build, `build/kg/`
`kg/data_dictionary.md`, `.json`	44 kB	local build, `build/kg/`; the `.md` file also from the downloads page
`rdf/vlonto-kg.nt.gz`	48.2 MB	local build, `build/rdf/`
`rdf/vlonto-core.ttl`	2.6 MB	local build, `build/rdf/`; downloads page
`ontology/vlonto.ttl`	122 kB	repository, `ontology/`; downloads page
`ontology/vlonto-shapes.ttl`	13 kB	repository, `ontology/`; downloads page
`ontology/vlonto-context.jsonld`	7 kB	repository, `ontology/`; downloads page
`ontology/vlonto.html`	680 kB	repository, `ontology/`; the website's ontology page

Sizes are those of the reference build of 2026-11-12 (section 10.4). `rdf/vlonto-kg.nt.gz` is the whole graph as gzip-compressed N-Triples; `rdf/vlonto-core.ttl` is the subset the browser loads (section 9); `ontology/` holds the OWL ontology, its SHACL shapes, a JSON-LD context and the HTML documentation; `kg/` holds the tables documented in Appendix C. The downloads page also offers the website's JSON data, the licence policy, the attribution text, the figures and this playbook as PDF and Markdown. The local-only table `article_text` (436,309 rows of article text) is written for local use only and is never shared (Appendix D).

2.3 Install the query libraries

The examples use the versions of the build environment (Python 3.11.18):

```
pip install duckdb==2.0.0 polars==2.1.0 pyoxigraph==0.5.13
```

Run the examples in the build folder of a local build (`build/`, or `$VLONTO_BUILD`), which contains `kg/` and `rdf/`. The SPARQL example of section 2.6 needs only `rdf/vlonto-core.ttl`, which the website's downloads page also offers.

2.4 SQL over the parquet tables with DuckDB

DuckDB reads parquet files in place; a table is its file name. Count documents per validity class (`playbook/examples/quickstart_duckdb.py`):

```
"""Count documents per validity class with DuckDB (run next to the kg/ folder)."""
import duckdb

con = duckdb.connect()
rows = con.execute("""
    SELECT validity, count(*) AS documents
    FROM 'kg/documents.parquet'
    GROUP BY validity
    ORDER BY documents DESC, validity
""").fetchall()
for validity, n in rows:
    print(f"{validity:<27}{n:>7,}")
```

Output:

```
unknown                     28,660
superseded                   5,167
presumed_in_force_amended    2,722
expired                      2,125
in_force                     1,511
in_force_amended               545
not_applicable                  47
not_yet_effective               39
```

2.5 DataFrames with polars

Find the Labour Code by its number and list its first articles (`playbook/examples/quickstart_polars.py`):

```
"""Find a document by its number and list its first articles with polars."""
import polars as pl

docs = pl.read_parquet("kg/documents.parquet")
code = docs.filter(pl.col("doc_number") == "45/2019/QH14")
for row in code.select("doc_id", "title_vi", "issue_date", "validity").iter_rows():
    print(*row, sep=" | ")

articles = (
    pl.scan_parquet("kg/articles.parquet")
    .filter(pl.col("doc_id") == code["doc_id"][0])
    .sort("position")
    .select("article_no", "heading")
    .head(4)
    .collect()
)
for no, heading in articles.iter_rows():
    print(f"Điều {no}: {heading}")
```

Output:

```
333670 | Bộ luật Lao động | 2019-11-20 | in_force_amended
Điều 1: Phạm vi điều chỉnh
Điều 2: Đối tượng áp dụng
Điều 3: Giải thích từ ngữ
Điều 4: Chính sách của Nhà nước về lao động
```

2.6 SPARQL with pyoxigraph

The browser subset loads into memory in a moment. List the codes (*bộ luật*) that are in force (`playbook/examples/quickstart_sparql.py`):

```
"""Load the browser-sized Turtle subset into pyoxigraph and run SPARQL."""
from pyoxigraph import RdfFormat, Store
```

```
store = Store()  # in memory
store.load(path="rdf/vlonto-core.ttl", format=RdfFormat.TURTLE)
print(f"{len(store):,} triples")

QUERY = """
PREFIX vlo: <https://tmquan.github.io/VLOnto/ontology#>
SELECT ?number ?title WHERE {
  ?code a vlo:Code ; vlo:title ?title ; vlo:isInForce true .
  OPTIONAL { ?code vlo:documentNumber ?number }
}
ORDER BY ?number
"""
for row in store.query(QUERY):
    number = row["number"].value if row["number"] else "-"
    print(number, row["title"].value)
```

Output:

```
49,846 triples
100/2015/QH13 Bộ luật Hình sự
101/2015/QH13 Bộ luật Tố tụng hình sự
45/2019/QH14 Bộ luật Lao động
74/2006/QH11 Bộ luật 74/2006/QH11
84/2007/QH11 Bộ luật 84/2007/QH11
91/2015/QH13 Bộ luật Dân sự
92/2015/QH13 Bộ luật Tố tụng dân sự
95/2015/QH13 Bộ luật Hàng hải Việt Nam
```

The subset holds the ontology and the 1,781 documents of the highest legal-force ranks (constitution; codes, laws and national assembly resolutions; ordinances and standing committee resolutions) with the 612 change events, 4,997 legal-basis edges and 6,198 document-level citations among them. Questions about decrees, circulars, Q&As or terms need the full dump, which is bulk-loaded into an on-disk store once (`playbook/examples/load_full_dump.py`):

```
"""Bulk-load the full N-Triples dump into an on-disk pyoxigraph store."""
import gzip

from pyoxigraph import RdfFormat, Store

store = Store("vlonto-store")  # RocksDB directory, created if missing
with gzip.open("rdf/vlonto-kg.nt.gz", "rb") as dump:
    store.bulk_load(dump, format=RdfFormat.N_TRIPLES)
store.optimize()
print(f"{len(store):,} triples in vlonto-store/")
```

Output:

```
6,963,753 triples in vlonto-store/
```

The build loaded the same dump into Oxigraph in 8.0 s (6,963,753 triples). Afterwards open the store with `Store("vlonto-store")` and query it as above; section 7 runs every SPARQL recipe against this store.

3. Architecture

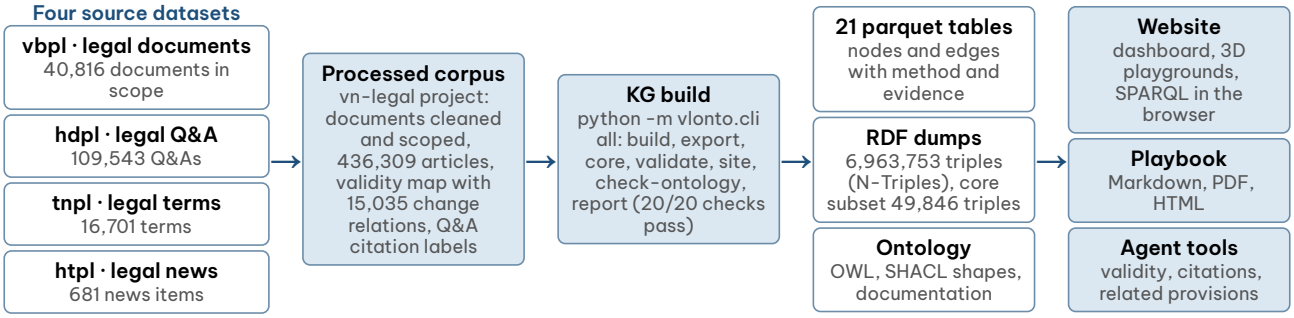


Figure 1: Data flow: the four source datasets, the processed corpus of the vn-legal project, the knowledge-graph build, the tables and RDF dumps it writes, and their uses.

Source datasets. Four datasets published on Hugging Face by `tmquan` are the data home of VLOnto v1; each is at `https://huggingface.co/datasets/tmquan/<dataset>` (the names in the table are links), pinned to a fixed revision:

Dataset	Content	Source	Revision	Licence
`thuvienphapluat-vn-hdpl`	legal Q&A	thuvienphapluat.vn	`e814ecbfa1a1`	other
`thuvienphapluat-vn-htpl`	legal news	thuvienphapluat.vn	`043c8f2988e2`	other
`thuvienphapluat-vn-tnpl`	legal terminology	thuvienphapluat.vn/tnpl	`f94d1e24852f`	other
`thuvienphapluat-vn-vbpl`	legal documents	thuvienphapluat.vn	`705ade585384`	cc-by-4.0

The court datasets `anle-toaan-gov-vn` and `cbba-toaan-gov-vn` are not read by VLOnto v1; they join in VLOnto v2.

Processed corpus. The vn-legal project (the data pipeline of the Vietnamese legal assistant, section 8) downloads the pinned revisions, cleans and scopes the documents, splits them into articles and extracts the validity map from the text of newer documents (validity rules v7). VLOnto reads its outputs instead of re-parsing the sources:

Input	File (vn-legal root R)
`chunks`	`$R/data/chunks/chunks.parquet`
`doc_struct`	`$R/data/chunks/doc_struct.parquet`
`docs_clean`	`$R/data/clean/docs_clean.parquet`
`gold_all`	`$R/eval/gold_all.parquet`
`relations`	`$R/data/validity/relations.parquet`
`validity`	`$R/data/validity/validity.parquet`

From the vbpl snapshot, 40,816 documents are in scope (the normative documents kept by the cleaning step of vn-legal); the Q&As, terms and news are read from the source snapshots directly.

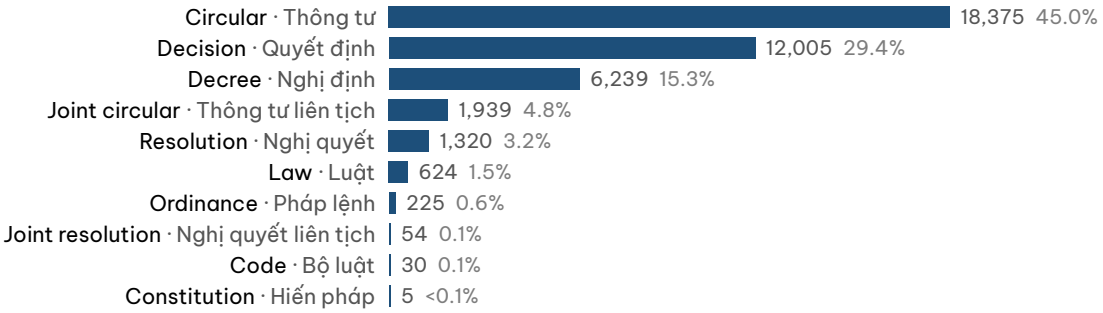


Figure 2: In-scope legal documents by type: number and share of the 40,816 documents.

Knowledge-graph build. `python -m vlonto.cli all` (section 5) resolves every reference to a document, article, clause or point, writes the node and edge tables with the method and evidence of each edge, exports RDF, validates, and writes the website data. 850,703 nodes and 556,287 edges are stored in the parquet tables.

Outputs. The parquet tables are the primary form (fast, columnar, with provenance columns); the RDF dump states the same facts with the ontology's IRIs (6,963,753 triples); the core Turtle subset serves the browser; the website and this playbook are generated from the build reports. There is no separate data release: the tables and the full dump are rebuilt from the source datasets, while the GitHub repository and the website carry the code, the ontology, the core subset and this playbook (section 2.2).

What is deliberately left out. No article or document bodies, no Q&A or news text, no glossary definitions and no person names are published; titles, article headings (up to 200 characters) and one-sentence evidence quotes (up to 300 characters) are kept as labels. Text stays at the source URL that every record carries (Appendix D). The website's question traces page (section 9) is the one exception: it shows the published title of each question it traces, with a short English gloss, but not the full question and no answer text.

4. The ontology

VLOnto is an OWL 2 ontology in Turtle (`ontology/vlonto.ttl`, 2,061 triples) with 47 classes, 46 object properties, 35 datatype properties, 3 annotation properties and 13 SKOS concept schemes. Every VLOnto term has an English and a Vietnamese label and an English definition (the completeness check of section 6), and 47 of the 131 terms also have a Vietnamese definition. Namespace: `https://tmquan.github.io/VLOnto/ontology#` (prefix `vlo:`); instances: `https://tmquan.github.io/VLOnto/id/` (prefix `vlid:`).

4.1 Modules

Each term belongs to one of 11 thematic modules (annotation `vlo:module`); the 3D ontology playground colours terms by module.

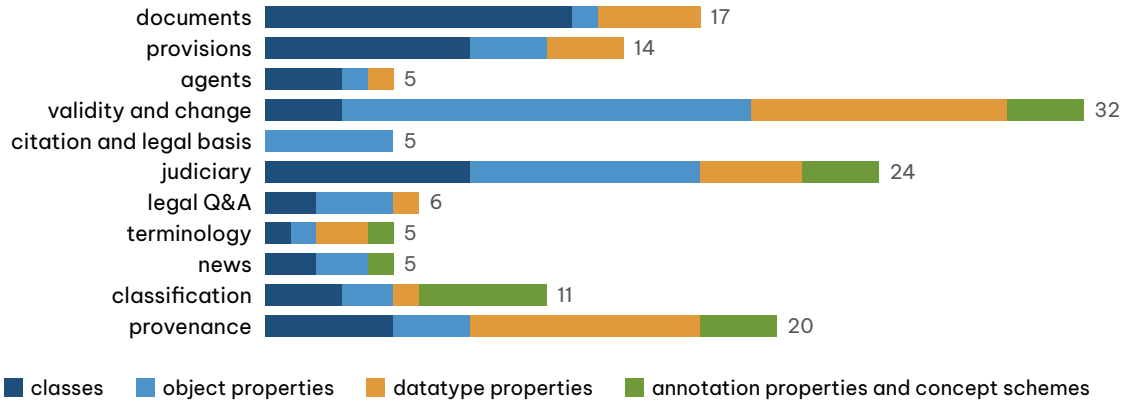


Figure 3: Number of VLOnto terms per module, by kind.

Module	Vietnamese	Scope	Classes	Object prop.	Data prop.
`documents`	văn bản	Legal documents and their types.	12	1	4
`provisions`	điều khoản	Structural units of legal documents: parts, chapters, sections, articles, clauses, points and appendices.	8	3	3
`agents`	chủ thể	Public authorities that issue documents or decide cases.	3	1	1
`validity-change`	hiệu lực và thay đổi	Validity status of documents and the dated events that replace, amend, repeal or terminate them.	3	16	10
`citation-basis`	trích dẫn và căn cứ	Legal-basis and citation links between legal resources and from other works to legal resources.	0	5	0
`judiciary`	tư pháp	Court decisions, judgments, precedents and their classification. No data in VLOnto v1; populated from v2 (section 4.2).	8	9	4
`qa`	hỏi đáp pháp luật	Published legal questions and their answers.	2	3	1
`terminology`	thuật ngữ	Legal terms organised as SKOS concepts.	1	1	2
`news`	tin tức	Legal news and policy articles that mention legal documents.	2	2	0
`classification`	phân loại	Concept schemes for legal areas, document types and legal-force ranks.	3	2	1
`provenance`	nguồn gốc	Source datasets, extraction methods, edge sets and build activities.	5	3	9

4.2 Key classes

Class	Vietnamese	Definition and superclasses
vlo:LegalDocument	Văn bản pháp luật	A document issued by a competent state body that has legal effect, identified by its type, number and issue date. Subclass of eli:LegalResource, schema:Legislation.
vlo:NormativeLegalDocument	Văn bản quy phạm pháp luật	A legal document containing generally binding rules of conduct, issued in the form, order and procedure set out by the Law on Promulgation of Legal Normative Documents. Subclass of vlo:LegalDocument.
vlo:Provision	Đơn vị cấu trúc của văn bản	A structural unit of a legal document, such as a part, chapter, section, article, clause, point or appendix. Subclass of eli:LegalResourceSubdivision.
vlo:Article	Điều	The basic numbered unit of a legal document (Điều N), usually with a heading, divided into clauses. Subclass of vlo:Provision.
vlo:Clause	Khoản	A numbered subdivision of an article (khoản 1, khoản 2, …). Subclass of vlo:Provision.
vlo:Point	Điểm	A lettered subdivision of a clause (điểm a, điểm b, …). Subclass of vlo:Provision.
vlo:IssuingBody	Cơ quan ban hành	A public authority that issues legal documents, identified in the knowledge graph by the issuer code used in document numbers (for example BTC for the Ministry of Finance). Subclass of vlo:PublicAuthority.
vlo:LegalChange	Sự kiện thay đổi hiệu lực văn bản	A dated event in which an acting legal document replaces, amends, repeals, partially repeals or partially terminates an affected legal document. The event is reified so that its kind, effective date, the evidence sentence and the extraction method can be recorded; the direct shortcut properties (vlo:replaces, vlo:amends, …) carry the same information without the details. Subclass of prov:Activity.
vlo:ValidityStatus	Tình trạng hiệu lực	The validity of a legal document as of the knowledge-graph date, combining the curated source status with text-extracted change relations (validity rules v7). It refines the ELI in-force concept with eight values. Subclass of eli:InForce, skos:Concept.
vlo:LegalQuestion	Câu hỏi pháp luật	A published legal question from a legal Q&A service. The knowledge graph keeps its identifier, link, date and legal area, not its text. Subclass of schema:Question.
vlo:LegalAnswer	Câu trả lời pháp luật	The published answer to a legal question; in the knowledge graph it carries the citations of legal documents and provisions found in the answer, not its text. Subclass of schema:Answer.
vlo:LegalTerm	Thuật ngữ pháp lý	A term of Vietnamese legal language, organised as a SKOS concept with a Vietnamese preferred label, an English label (machine translation unless marked otherwise), a legal area, a status and related terms. Subclass of skos:Concept.
vlo:LegalArea	Lĩnh vực pháp luật	A subject area of law used to classify documents, questions and terms, such as labour and wages or real estate. Subclass of skos:Concept.
vlo:SourceDataset	Bộ dữ liệu nguồn	A published dataset, pinned to a revision, from which knowledge-graph records were derived; carries its licence and attribution text. Subclass of prov:Entity.
vlo:EdgeSet	Tập liên kết	The set of all knowledge-graph links of one property produced from one source, described with its size, the extraction methods used and the build that generated it. Per-link methods are kept in the parquet edge tables. Subclass of prov:Entity.

Documents are typed by 10 subclasses of vlo:NormativeLegalDocument, one per document type, each linked to its concept in the document-type scheme: vlo:Constitution (*Hiến pháp*), vlo:Code (*Bộ luật*), vlo:Law (*Luật*), vlo:Ordinance (*Pháp lệnh*), vlo:Resolution (*Nghị quyết*), vlo:JointResolution (*Nghị quyết liên tịch*), vlo:Decree (*Nghị định*), vlo:Decision (*Quyết định*), vlo:Circular (*Thông tư*), vlo:JointCircular (*Thông tư liên tịch*). Structural units: vlo:Part, vlo:Chapter, vlo:Section and vlo:Appendix.

Judiciary terms (v2). The judiciary module keeps its 21 terms in the ontology, among them vlo:CourtDecision, vlo:Judgment, vlo:CourtRuling, vlo:Precedent, and the properties vlo:decidedBy, vlo:appliesProvision and vlo:citesCase (Appendix B.6). They, the module's concept schemes and the class vlo:Court carry a scope note in English and Vietnamese (26 terms, schemes and modules), which

says that they are populated from VLOnto v2; the module's note reads: *The terms of this module are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no court data.* (*Các khái niệm của mô-đun này có dữ liệu từ VLOnto 2.0 (các bộ dữ liệu tòa án anle-toaan-gov-vn và cbba-toaan-gov-vn); đồ thị tri thức VLOnto 1.0.0 chưa có dữ liệu tòa án.*)

4.3 Key properties

Property	Domain	Range	Sub-property of
vlo:legalBasis	vlo:LegalDocument	vlo:LegalDocument	eli:based_on, vlo:refersTo
vlo:cites	eli:LegalResource	eli:LegalResource	eli:cites, vlo:refersTo
vlo:citesParentLaw	eli:LegalResource	eli:LegalResource	vlo:cites
vlo:replaces	vlo:LegalDocument	vlo:LegalDocument	eli:repeals, vlo:changes
vlo:amends	vlo:LegalDocument	vlo:LegalDocument	eli:amends, vlo:changes
vlo:repeals	vlo:LegalDocument	vlo:LegalDocument	eli:repeals, vlo:changes
vlo:partiallyRepeals	vlo:LegalDocument	vlo:LegalDocument	vlo:changes
vlo:partiallyTerminates	vlo:LegalDocument	vlo:LegalDocument	vlo:changes
vlo:actingDocument	vlo:LegalChange	vlo:LegalDocument	-
vlo:affectedDocument	vlo:LegalChange	vlo:LegalDocument	-
vlo:changeKind	vlo:LegalChange	vlo:ChangeKind	-
vlo:validityStatus	vlo:LegalDocument	vlo:ValidityStatus	eli:in_force
vlo:isPartOf	vlo:Provision	eli:LegalResource	eli:is_part_of
vlo:inDocument	vlo:Provision	vlo:LegalDocument	eli:is_part_of
vlo:issuedBy	vlo:LegalDocument	vlo:IssuingBody	eli:passed_by
vlo:answerCitesProvision	vlo:LegalAnswer	vlo:Provision	schema:citation, vlo:refersTo
vlo:answerCitesDocument	vlo:LegalAnswer	vlo:LegalDocument	schema:citation, vlo:refersTo
vlo:relatedTerm	vlo:LegalTerm	vlo:LegalTerm	skos:related
vlo:sourceDataset	-	vlo:SourceDataset	dcterms:source

A change has two representations. The event (`vlo:LegalChange`) records the acting and the affected document, the kind, the effective date, the evidence sentence and the rule version; the shortcut triple (`vlid:document/A vlo:replaces vlid:document/B`) makes simple queries short. Citations point to the most specific target that was resolved: a point, else a clause, else an article, else the document; `vlo:isPartOf` links a point to its clause, a clause to its article and an article to its chapter or document, and `vlo:inDocument` links every provision directly to its document.

4.4 Alignments

VLOnto reuses established vocabularies through sub-class and sub-property axioms rather than redefining them: 62 alignment axioms to 42 external terms in 6 vocabularies. A tool that knows ELI, SKOS, PROV-O, schema.org, ORG or DCTERMS can therefore read VLOnto data after reasoning (the OWL-RL closure of the validation sample infers 4,752 `eli:LegalResource` instances, section 6).

Vocabulary	Axioms	Examples (VLOnto term → term it specialises)
DCMI Terms	12	vlo:heading → dcterms:title, vlo:legalArea → dcterms:subject, vlo:provisionNumber → dcterms:identifier, …
ELI - European Legislation Identifier	20	vlo:DocumentType → eli:ResourceType, vlo:LegalDocument → eli:LegalResource, vlo:Provision → eli:LegalResourceSubdivision, …
W3C Organization ontology	1	vlo:PublicAuthority → org:Organization
PROV-O	6	vlo:BuildActivity → prov:Activity, vlo:EdgeSet → prov:Entity, vlo:ExtractionMethod → prov:Plan, …
schema.org	10	vlo:LegalAnswer → schema:Answer, vlo:LegalDocument → schema:Legislation, vlo:LegalNewsArticle → schema:NewsArticle, …
SKOS	13	vlo:ChangeKind → skos:Concept, vlo:DocumentType → skos:Concept, vlo:ExtractionMethod → skos:Concept, …

The ELI terms used were checked against the ELI ontology itself: all 18 exist there (124 ELI terms defined). Appendix B lists every axiom with its term.

4.5 IRI scheme

Instance IRIs are built from the source identifiers, so they are stable across rebuilds of the same source revisions:

Node	IRI pattern	Note
document	`vlid:document/<doc_id>`	vbpl document id
article	`vlid:document/<doc_id>/dieu-<n>`	appendix articles: `att<s>-dieu-<n>`
clause, point	`.../dieu-<n>/khoan-<k>/diem-<p>`	created when an edge refers to them
part, chapter, section, appendix	`vlid:document/<doc_id>/chuong-iii/muc-3`	structure path
change event	`vlid:change/<acting>-<kind>-<affected>`	one per relation
Q&A, answer	`vlid:qa/<id>`, `vlid:qa/<id>/answer`	hdpl id
term, news item	`vlid:term/<id>`, `vlid:news/<id>`	tnpl and htpl ids
area, issuer	`vlid:area/<slug>`, `vlid:issuer/<code>`	curated codes
source, edge set, build	`vlid:source/<dataset>`, `vlid:edgeset/<name>`	provenance nodes

The namespaces are under the website address `https://tmquan.github.io/VLOnto/`. GitHub Pages serves static files without content negotiation, so a client that dereferences an IRI gets a web page or nothing, not RDF: load the Turtle file or the dump instead of dereferencing IRIs.

4.6 Versioning

The ontology header carries `owl:versionInfo "1.0.0"`, `owl:versionIRI <https://tmquan.github.io/VLOnto/ontology/1.0.0>`, `owl:priorVersion <https://tmquan.github.io/VLOnto/ontology/0.1.0>`, `dcterms:created` and `dcterms:modified` (2026-11-12) and the licence `<https://creativecommons.org/licenses/by/4.0/>`. Release v1 is version 1.0.0 of the ontology and of the build. Every knowledge-graph build records the same version and its validity date (`vlo:softwareVersion` and `vlo:validityAsOf` on the build activity node). Proposed release policy for the repository, semantic versioning: a patch release changes labels, definitions or data only; a minor release adds classes or properties; a major release renames or removes terms or changes the IRI scheme. Each release keeps its version IRI, so data can state which version it conforms to.

5. Building the knowledge graph

5.1 Prerequisites

The build runs on one machine with the processed corpus of the vn-legal project (root `R`) and the pinned source snapshots in its Hugging Face cache. It needs a Python environment with duckdb 2.0.0, igraph 1.0.0, owlrl 7.6.2, polars 2.1.0, pyarrow 26.0.1, pylode 3.6.1, pyoxigraph 0.5.13, pyshacl 0.40.2, rdflib 7.6.0; these are the versions of the reference build of VLOnto v1, which ran with Python 3.11.18 on macOS, arm64 (`scripts/build_pdf.sh` looks for the environment at `$V/envs/kg`). Paths default to the values in `vlonto/vlonto/config.py` and can be overridden with the environment variables `R` (vn-legal root), `V` (VLOnto root), `VLONTO_REPO` and `VLONTO_BUILD`.

5.2 Commands

```
export R=/path/to/vnlegal V=/path/to/vlonto
cd $V/repo/vlonto
$V/envs/kg/bin/python -m vlonto.cli all            # every step below, in order
$V/envs/kg/bin/python -m vlonto.cli build          # or one step at a time
$V/envs/kg/bin/python -m vlonto.cli report         # regenerate kg_report.md only
```

Options: `--workers N` (worker processes of the RDF export), `--core-max-rank N` (legal-force ranks in the browser subset), `--no-article-text` (skip the local-only article text table) and `--with-article-text` (copy article text into the articles table; a licence decision before the tables are shared, Appendix D). `layout` recomputes only the 3D layout of the legal network. `--with-courts` (or `VLONTO_INCLUDE_COURTS=1`) also reads the court datasets; it is for the work on v2 and is not part of VLOnto v1.

5.3 Steps and expected output

Step	Writes	Expected result	Wall time
`build`	`build/kg/*.parquet`, `build/kg_local/`	40,816 documents, 436,309 articles, 15,035 change events, 556,287 edges	33 s
`export`	`build/rdf/vlonto-kg.nt.gz`	6,963,753 triples, 48.2 MB	7 s
`core`	`build/rdf/vlonto-core.ttl`	49,846 triples, 1,781 documents	0.9 s
`validate`	`build/reports/validation.json`, `build/oxigraph/`	32 integrity checks, 6,963,753 triples loaded	37 s
`site`	`build/site_data/*.json`, `js/*.js`	15 bundles and twins, all within 20 MB	-
`check-ontology`	`build/reports/ontology_check.json`, `ontology/vlonto.html`	131 terms checked, 971 documentation links	-
`report`	`build/reports/kg_report.md`, `checks.json`	20 of 20 checks pass	-

Wall times are those of the reference build (macOS, arm64, 2026-11-12); - means the step reports no time. The step reports are JSON files in `build/reports/`; `kg_report.md` collects them. A rebuild from the same source revisions must reproduce every count in this playbook; the reconciliation check (section 6.2) compares the main counts with the data report of the vn-legal project automatically.

5.4 Node and edge tables

Table	Rows	Content
`documents`	40,816	Legal documents (40,816 in-scope normative documents from vbpl, after cleaning).
`articles`	436,309	Articles (Điều): one node per main-body or appendix article; parts of split articles joined. No text (licence).
`structure`	73,925	Appendices, parts, chapters and sections of documents.
`provisions_ref`	48,031	Clauses (khoản) and points (điểm) created because an edge refers to them.
`issuers`	57	Issuing bodies identified by the symbol of the document number (curated table; unknown symbols get no node).
`areas`	51	Legal areas (SKOS concepts) from vbpl/hdpl (27) and tnpl.

Table	Rows	Content
doc_types	10	Document types (concepts of the ontology).
legal_force_ranks	6	Legal-force ranks (concepts of the ontology).
changes	15,035	Legal change events (validity relations v7), one per relation.
edges_change	15,035	Shortcut edges acting document -> affected document (one per change event).
edges_legal_basis	99,038	Legal basis: citing document -> basis document.
edges_citation	268,563	vbpl citations (external, external_mention, parent_law, parent_law_mention) resolved to documents/articles.
qas	109,543	Legal Q&As from hdpl: metadata only (no question or answer text).
edges_qa_article	34,869	Q&A answer -> article (+clause, point) from the gold resolver (gold_all.parquet).
edges_qa_document	119,683	Q&A answer -> document cited by number.
terms	16,701	Legal terms from tnpl (names only, no definitions).
edges_term_related	17,615	Related terms.
news	681	Legal news articles from htpl (no title or text).
edges_news_document	1,484	News article -> mentioned document.
news_categories	7	News categories.
sources	4	Source datasets of this build with revision, licence and policy (release v1: the four thuvienphapluat.vn datasets).

Every edge table keeps a `method` column (how the edge was found: `docnum-key`, `law-name-year`, `preamble-citation`, `answer-text-name` and so on) and, where the source has one, the raw reference as written, so that any edge can be traced and re-judged. The columns of every table are listed in Appendix C.

6. Validation

6.1 The 20 build checks

`vlonto.cli validate`, `check-ontology` and `report` run 20 checks; 20 pass in the build of 2026-11-12.

#	Check	Result	Evidence (as recorded by the build)
1	Ontology completeness (en+vi labels, en definition, module for every VLOnto term)	pass	`0 problems; terms by kind {'class': 47, 'object_property': 46, 'datatype_property': 35, 'annotation_property': 3}`
2	Ontology has no dangling references	pass	`dangling=[]`
3	Ontology release metadata (owl:versionInfo 1.0.0, owl:versionIRI https://tmquan.github.io/VLOnto/ontology/1.0.0, dcterms:modified = build date 2026-11-12)	pass	`versionInfo 1.0.0, versionIRI https://tmquan.github.io/VLOnto/ontology/1.0.0, priorVersion https://tmquan.github.io/VLOnto/ontology/0.1.0, modified 2026-11-12; dcterms:source 4 datasets (match: True); judiciary terms with a v2 scope note 26/26`
4	ELI terms used by VLOnto exist in the ELI ontology	pass	`124 ELI terms defined; missing []`
5	OWL-RL on fixtures (expected inferences present, planted disjointness and functional clash detected)	pass	`27 expected inferences, missing []; negative: 2 inconsistencies`
6	SHACL on fixtures (positive conforms, all planted errors reported)	pass	`positive results 0; planted found ['bad-article:inDocument', 'bad-article:provisionNumber', 'bad-change:actingDocument', 'bad-change:evidenceText', 'bad-doc:documentType', 'bad-term:definition'], missed []`
7	pyLODE documentation renders with no broken internal links	pass	`680464 bytes, 971 internal links, broken 0`
8	Referential integrity of every parquet edge and foreign key	pass	`32 checks, orphans 0`
9	Reconciliation with the processed-corpus data report	pass	`21/21 items match`
10	Full RDF dump parses strictly and loads into Oxigraph	pass	`6,963,753 N-Triples lines written, 6,963,753 distinct triples loaded`
11	Per-class instance counts identical in parquet and RDF	pass	`20/20 classes match`
12	Every VLOnto instance IRI used as an object is typed in the dump	pass	`312,524 object IRIs, 0 untyped`
13	SHACL on a stratified sample of the KG (up to 150 nodes per class or kind)	pass	`3,066 nodes / 42,857 triples; violations 0`
14	OWL-RL closure of the KG sample is consistent	pass	`25,681 -> 91,886 triples; inconsistencies 0, sameAs between distinct nodes 0`
15	Release v1 scope: no court data (no court table, no court/decision/precedent node, no court-source statement); sources = the 4 thuvienphapluat datasets	pass	`court tables present []; typed court nodes in RDF {'Court': 0, 'CourtDecision': 0, 'Judgment': 0, 'CourtRuling': 0, 'Precedent': 0, 'JudicialDocument': 0}; dump lines mentioning court IRIs or court sources 0; source datasets ['thuvienphapluat-vn-vbpl', 'thuvienphapluat-vn-hdpl', 'thuvienphapluat-vn-tnpl', 'thuvienphapluat-vn-htpl']`
16	Spot checks: 10 random edges per edge type read and judged (criterion: at least 8 correct per 10 for every edge type)	pass	`127/130 judged correct, 1 judged wrong, 2 not verifiable from the record; edge types below the criterion: []`
17	Site bundles within 20 MB, each with a file:// .js twin	pass	`15 files, 9 js twins, over 20 MB: []`
18	0 non-NFC strings in every published output (Unicode NFC)	pass	`0 non-NFC strings in 47 files {'kg': 21, 'rdf': 2, 'site_data': 16, 'ontology': 4, 'docs': 4}; kg_report.md and checks.json are written in NFC by the report step`
19	No link to a VLOnto Hugging Face dataset or Space in any published output (release = GitHub repository + website)	pass	`49 files scanned; forbidden hits: links to a Hugging Face VLOnto dataset 0, links to a Space 0, Space path segments 0, wording that calls a VLOnto document a card 0; kept: 44 links to the tmquan source datasets on Hugging Face (4 sources of this build, 2 planned for v2), 7 to the GitHub repository, 7 to the paper (arXiv:2611.11426)`

#	Check	Result	Evidence (as recorded by the build)
20	No absolute build-machine path or host name in any published output (paths relative to the VLOnto root)	pass	`49 files scanned; build-node home paths 0, build-node host names 0, macOS home paths 0`

6.2 Reconciliation with the processed corpus

The build counts are compared item by item with the data report of the vn-legal project; all 21 of 21 items match:

Item	Data report	Knowledge graph	Match
documents	40,816	40,816	yes
change relations (total)	15,035	15,035	yes
change relations: replaced	6,321	6,321	yes
change relations: amended	4,678	4,678	yes
change relations: partially_repealed	1,854	1,854	yes
change relations: repealed	1,674	1,674	yes
change relations: partially_expired	508	508	yes
validity: unknown	28,660	28,660	yes
validity: superseded	5,167	5,167	yes
validity: presumed_in_force_amended	2,722	2,722	yes
validity: expired	2,125	2,125	yes
validity: in_force	1,511	1,511	yes
validity: in_force_amended	545	545	yes
validity: not_applicable	47	47	yes
validity: not_yet_effective	39	39	yes
main-body articles	425,892	425,892	yes
documents with main-body articles	32,902	32,902	yes
Q&As with gold article labels	22,523	22,523	yes
hdpl Q&As	109,543	109,543	yes
tnpl terms (rows minus not_found placeholders)	16,701	16,701	yes
htpl news	681	681	yes

6.3 Integrity, RDF and SHACL/OWL-RL

- **Referential integrity.** 32 foreign-key checks over every edge table; 0 orphans.
- **RDF.** The full dump parses strictly and loads into Oxigraph (6,963,753 distinct triples); per-class instance counts are identical in parquet and RDF for 20 classes; all 312,524 instance IRIs used as objects are typed (0 untyped).
- **SHACL.** `ontology/vlonto-shapes.ttl` has 18 node shapes with 90 property shapes. On fixtures, the positive file conforms and all 6 planted errors are reported. On a sample of the graph stratified by class and kind (3,066 nodes, 42,857 triples) there are 0 violations.
- **OWL-RL.** On fixtures, 27 expected inferences are all present and the planted disjointness and functional-property clashes are detected (2 inconsistencies). The closure of the graph sample grows from 25,681 to 91,886 triples with 0 inconsistencies and 0 `owl:sameAs` links between distinct nodes.
- **Ontology hygiene.** No dangling references; all 18 ELI terms used exist in the ELI ontology; the pyLODE documentation has 971 internal links, 0 broken.

6.4 Spot checks of extracted edges

For each edge type and resolution method, edges were drawn at random from the final build and judged by reading the numbers, titles, dates and the raw reference stored with the edge: 127 of 130 judged correct, 1 wrong and 2 not verifiable from the record; every edge type met the criterion of the spot-check entry in section 6.1. For VLOnto v1 the samples are the edges judged for the previous release, except the change events, which were drawn and judged again (`docs/spot_check_review.json`).

Edge type	Checked	Correct	Wrong	Not verifiable
`changes`	10	10	0	0
`legal_basis:parent-acts`	10	9	1	0
`legal_basis:preamble-citation`	10	9	0	1
`legal_basis:constitution-preamble`	10	10	0	0
`citation:external`	10	10	0	0
`citation:external_mention`	10	10	0	0
`citation:parent_law`	10	10	0	0
`citation:parent_law_mention`	10	10	0	0
`document_issuer`	10	10	0	0
`qa_article`	10	9	0	1
`qa_document`	10	10	0	0
`term_related`	10	10	0	0
`news_document`	10	10	0	0

The notes on the edges that were not judged correct:

- `legal_basis:parent-acts`: #7: the decree on administrative penalties for cybersecurity and personal data lists the Customs Law 54/2014/QH13 among its basis acts in the source metadata (source error carried over).
- `legal_basis:preamble-citation`: #9: Circular 18/2000/TT-BLĐTBXH mentions Circular 13/2000/TT-BLĐTBXH in its preamble; whether it is a legal basis or a cross-reference cannot be told from the record.
- `qa_article`: #1: ‘Điều 45 Khoản 3 Luật Đầu tư’ is linked to the PPP Investment Law 64/2020/QH14 (gold label of the evaluation set; the truncated reference string cannot settle it).

A sample of 10 gives only coarse precision: with 10 of 10 correct, the two-sided 95% lower confidence bound is about 69%. The samples catch systematic errors (several rules were tightened after earlier samples of the same day), not rare ones.

6.5 Precision of the validity relations

The 15,035 change events come from the validity map of the vn-legal project (rules v7). On a fresh sample of 30 relations, judged from the evidence sentence, 29 were correct (replaced or repealed 15/15, amended 10/10, partial 4/5; Wilson 95% interval 0.833 to 0.994), and all 3 known replacements were found:

Old act	Replaced by	From
10/2012/QH13	45/2019/QH14	2021-01-01
45/2013/QH13	31/2024/QH15	2024-08-01
58/2014/QH13	41/2024/QH15	2025-07-01

These judgements were made by the build agent from the evidence sentences, not by a lawyer.

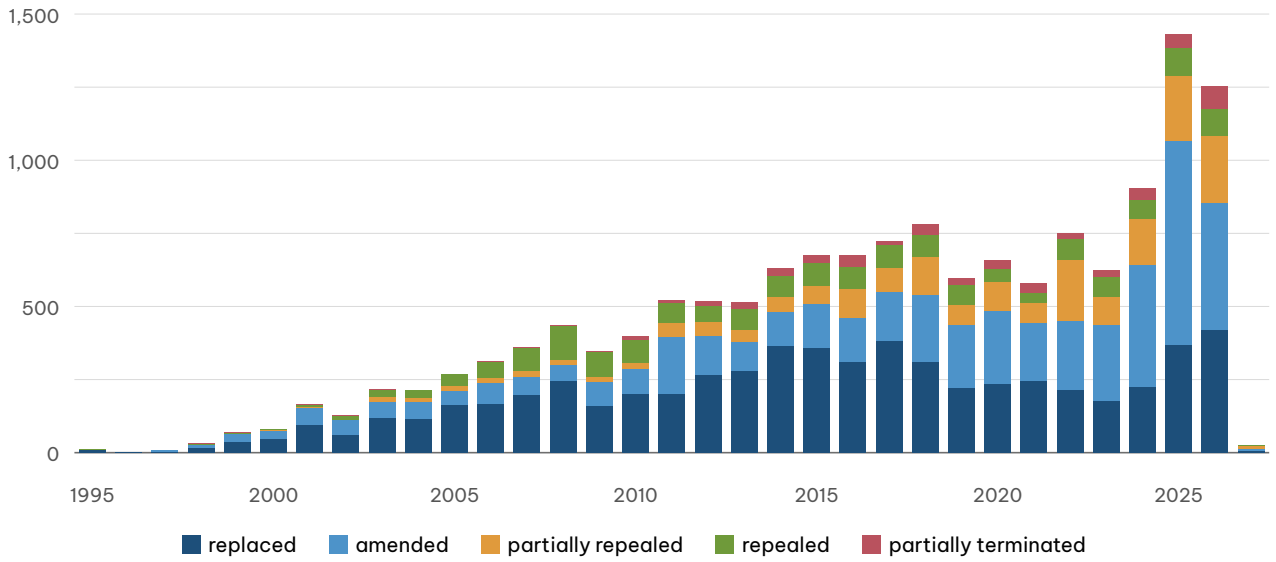


Figure 4: Change events per year of their effective date, by kind, 1995 to 2027: 14,903 of the 15,035 events; 99 earlier events and 33 events without an effective date are not shown.

6.6 Known limits of the validation

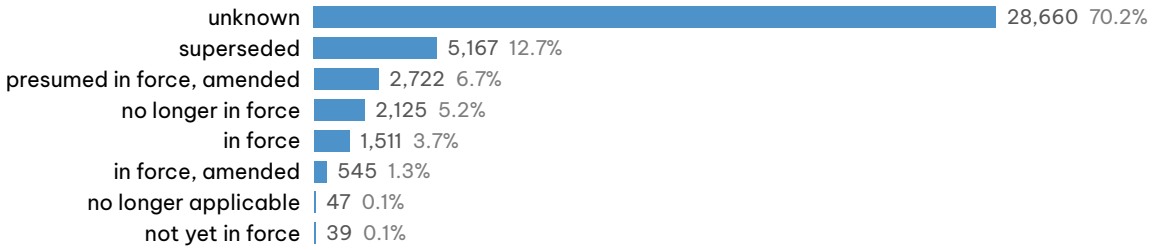


Figure 5: Validity classes of the 40,816 documents as of 2026-11-08 (validity rules v7).

- Validity is unknown for 28,660 documents (70.2%): the source has no status for them and no relation names them. Unknown is not “in force”.
- Precision figures rest on small samples; recall is not measured for most edge types.
- SHACL and OWL-RL run on samples and fixtures; the full graph is validated by the integrity and RDF checks.

The complete list of limitations is in section 12.

7. Recipes

Each recipe answers one question twice: in SQL over the parquet tables (DuckDB) and in SPARQL over the RDF. Every query below was run against the build of 2026-11-12; the outputs are printed as they were returned (long results are cut to their first rows). The SQL runs in the folder that contains `kg/`; the SPARQL runs on the full dump loaded as in section 2.6, and the row count on the browser subset is given where the subset can answer.

7.1 What replaced an act?

The Labour Code 10/2012/QH13 was replaced; the change event keeps the operative sentence of the new code as evidence.

SQL (`playbook/recipes/r1_replaced_by.sql`):

```
-- What replaced (or repealed) the Labour Code 10/2012/QH13?
SELECT n.doc_number AS new_number,
       n.title_vi   AS new_title,
       c.kind,
       c.effective_date
FROM 'kg/changes.parquet'   AS c
JOIN 'kg/documents.parquet' AS o ON o.doc_id = c.affected_doc_id
JOIN 'kg/documents.parquet' AS n ON n.doc_id = c.acting_doc_id
WHERE o.doc_number = '10/2012/QH13'
  AND c.kind IN ('replaced', 'repealed')
ORDER BY c.effective_date, new_number;
```

Output (1 row):

new_number	new_title	kind	effective_date
45/2019/QH14	Bộ luật Lao động	replaced	2021-01-01

SPARQL (`playbook/recipes/r1_replaced_by.rq`):

```
PREFIX vlo: <https://tmquan.github.io/VLOnto/ontology#>
# What replaced the Labour Code 10/2012/QH13?
SELECT ?number ?date ?evidence WHERE {
  ?old vlo:documentNumber "10/2012/QH13" .
  ?change vlo:affectedDocument ?old ;
          vlo:changeKind vlo:change-replaced ;
          vlo:actingDocument ?new ;
          vlo:evidenceText ?evidence .
  OPTIONAL { ?change vlo:changeEffectiveDate ?date }
  ?new vlo:documentNumber ?number .
}
ORDER BY ?date ?number
```

Output on the full dump (1 row; on the browser subset: 1 row):

number	date	evidence
45/2019/QH14	2021-01-01	Bộ luật Lao động số 10/2012/QH13 hết hiệu lực thi hành kể từ ngày Bộ luật này có hiệu lực.

Use the kinds `replaced` and `repealed` for “no longer in force because of X”, and `amended`, `partially_repealed` and `partially_expired` for “still in force, but changed” (recipe 7.2). The validity class of the affected document (`documents.validity`) combines these events with the status in the source.

7.2 Amendment history

All change events that affect the Land Law 31/2024/QH15, oldest first:

SQL (`playbook/recipes/r2_amendment_history.sql`):

```
-- Amendment history of the Land Law 31/2024/QH15: every change event
-- whose affected document is the law, oldest first.
SELECT c.effective_date,
```

```
       a.doc_number AS acting_number,
       a.doc_type_vi AS acting_type,
       c.kind
FROM 'kg/changes.parquet'   AS c
JOIN 'kg/documents.parquet' AS d ON d.doc_id = c.affected_doc_id
JOIN 'kg/documents.parquet' AS a ON a.doc_id = c.acting_doc_id
WHERE d.doc_number = '31/2024/QH15'
ORDER BY c.effective_date NULLS LAST, acting_number, c.kind;
```

Output (8 rows):

effective_date	acting_number	acting_type	kind
2024-08-01	43/2024/QH15	Luật	amended
2025-01-01	58/2024/QH15	Luật	amended
2025-07-01	47/2024/QH15	Luật	amended
2025-07-01	84/2025/QH15	Luật	partially_repealed
2026-01-01	146/2025/QH15	Luật	partially_repealed
2026-01-01	147/2025/QH15	Luật	amended
2026-01-01	95/2025/QH15	Luật	amended
2026-07-01	130/2025/QH15	Luật	amended

SPARQL (`playbook/recipes/r2_amendment_history.rq`):

```
PREFIX vlo:  <https://tmquan.github.io/VLOnto/ontology#>
PREFIX skos: <http://www.w3.org/2004/02/skos/core#>
# Amendment history of the Land Law 31/2024/QH15
SELECT ?date ?number ?kind WHERE {
  ?law vlo:documentNumber "31/2024/QH15" .
  ?change vlo:affectedDocument ?law ;
          vlo:actingDocument ?by ;
          vlo:changeKind ?k .
  ?k skos:prefLabel ?kind FILTER (lang(?kind) = "en")
  OPTIONAL { ?change vlo:changeEffectiveDate ?date }
  ?by vlo:documentNumber ?number .
}
ORDER BY ?date ?number ?kind
```

Output on the full dump (8 rows; on the browser subset: 8 rows):

date	number	kind
2024-08-01	43/2024/QH15	amended
2025-01-01	58/2024/QH15	amended
2025-07-01	47/2024/QH15	amended
2025-07-01	84/2025/QH15	partially repealed
2026-01-01	146/2025/QH15	partially repealed
2026-01-01	147/2025/QH15	amended
2026-01-01	95/2025/QH15	amended
2026-07-01	130/2025/QH15	amended

7.3 Legal-basis chain

A circular rests on decrees and laws, which rest on the Constitution. The SQL follows `edges_legal_basis` up to three steps and keeps the shortest depth of each document; the SPARQL property path `vlo:legalBasis+` follows the chain to any depth.

SQL (`playbook/recipes/r3_legal_basis_chain.sql`):

```
-- Legal-basis chain of the enterprise accounting circular 99/2025/TT-BTC:
-- follow vlo:legalBasis edges up to three steps, keep the shortest depth.
WITH RECURSIVE chain(doc_id, depth, path) AS (
    SELECT doc_id, 0, [doc_id]
    FROM 'kg/documents.parquet'
    WHERE doc_number = '99/2025/TT-BTC'
  UNION ALL
    SELECT e.dst_doc_id, c.depth + 1, list_append(c.path, e.dst_doc_id)
```

```
    FROM chain AS c
    JOIN 'kg/edges_legal_basis.parquet' AS e ON e.src_doc_id = c.doc_id
    WHERE c.depth < 3 AND NOT list_contains(c.path, e.dst_doc_id)
)
SELECT min(c.depth) AS depth,
       coalesce(d.doc_number, '-') AS number,
       d.doc_type_vi AS type,
       d.title_vi AS title
FROM chain AS c
JOIN 'kg/documents.parquet' AS d USING (doc_id)
WHERE c.depth > 0
GROUP BY d.doc_number, d.doc_type_vi, d.title_vi
ORDER BY depth, number, title;
```

Output (first 12 of 16 rows):

depth	number	type	title
1	04/2007/QH12	Luật	Luật Thuế thu nhập cá nhân
1	15/2012/QH13	Luật	Luật Xử lý vi phạm hành chính
1	166/2025/NĐ-CP	Nghị định	Nghị định 166/2025/NĐ-CP sửa đổi, bổ sung một số điều của nghị định số 29/2025/nđ-cp ngày 24 tháng 02 năm 2025 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của bộ tài chính
1	22/2012/QH13	Luật	Luật Dự trữ quốc gia
1	29/2025/NĐ-CP	Nghị định	Nghị định 29/2025/NĐ-CP quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của bộ tài chính
1	38/2019/QH14	Luật	Luật Quản lý thuế
1	54/2019/QH14	Luật	Luật Chứng khoán
1	67/2011/QH12	Luật	Luật Kiểm toán độc lập
1	88/2015/QH13	Luật	Luật Kế toán
1	89/2025/QH15	Luật	Luật Ngân sách Nhà nước
2	-	Hiến pháp	Hiến pháp Nước cộng hoà chủ nghiã xã hội Việt Nam năm 1992
2	-	Hiến pháp	Hiến pháp Nước cộng hòa xã hội chủ nghĩa Việt Nam năm 2013

SPARQL (`playbook/recipes/r3_legal_basis_chain.rq`):

```
PREFIX vlo: <https://tmquan.github.io/VLOnto/ontology#>
# Every document reachable from 99/2025/TT-BTC by vlo:legalBasis (any depth)
SELECT DISTINCT ?number ?title WHERE {
  ?doc vlo:documentNumber "99/2025/TT-BTC" ;
       vlo:legalBasis+ ?basis .
  ?basis vlo:title ?title .
  OPTIONAL { ?basis vlo:documentNumber ?number }
}
ORDER BY ?number ?title
```

Output on the full dump (first 8 of 16 rows; on the browser subset: 0 rows):

number	title
-	Hiến pháp Nước cộng hoà chủ nghiã xã hội Việt Nam năm 1992
-	Hiến pháp Nước cộng hòa xã hội chủ nghĩa Việt Nam năm 2013
04/2007/QH12	Luật Thuế thu nhập cá nhân
15/2012/QH13	Luật Xử lý vi phạm hành chính
166/2025/NĐ-CP	Nghị định 166/2025/NĐ-CP sửa đổi, bổ sung một số điều của nghị định số 29/2025/nđ-cp ngày 24 tháng 02 năm 2025 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của bộ tài chính
194/2025/QH15	Nghị quyết 194/2025/QH15 về việc sửa đổi, bổ sung một số điều của hiến pháp năm 2013
203/2025/QH15	Nghị quyết 203/2025/QH15 sửa đổi, bổ sung một số điều của hiến pháp nước cộng hòa xã hội chủ nghĩa Việt Nam
22/2012/QH13	Luật Dự trữ quốc gia

The browser subset returns no row here because a circular is outside it (section 2.6). Amending resolutions of the Constitution appear because the laws in the chain cite them as their basis.

7.4 Articles most cited in Q&A answers

Q&A article edges link a published legal Q&A to the article, clause or point that its answer cites (the evaluation labels of the vn-legal project, section 7.5). Counting Q&As per article:

SQL (`playbook/recipes/r4_qa_articles.sql`):

```
-- Articles cited by the largest number of published Q&A answers
-- (a Q&A whose answer cites several clauses of one article counts once).
SELECT d.doc_number,
       a.article_no,
       a.heading,
       count(DISTINCT e.qa_id) AS qas
FROM 'kg/edges_qa_article.parquet' AS e
JOIN 'kg/articles.parquet'  AS a ON a.article_id = e.dst_article_id
JOIN 'kg/documents.parquet' AS d ON d.doc_id = a.doc_id
GROUP BY a.article_id, d.doc_number, a.article_no, a.heading
ORDER BY qas DESC, d.doc_number, a.article_no
LIMIT 10;
```

Output (10 rows):

doc_number	article_no	heading	qas
45/2019/QH14	111	Nghỉ hằng tuần	166
12/2022/NĐ-CP	46	Vi phạm của người lao động Việt Nam đi làm việc ở nước ngoài và một số đối tượng liên quan khác	165
45/2019/QH14	112	Nghỉ lễ, tết	161
45/2019/QH14	169	Tuổi nghỉ hưu	155
04/2007/QH12	19	-	139
65/2013/NĐ-CP	3	Thu nhập chịu thuế	130
38/2019/QH14	44	Thời hạn nộp hồ sơ khai thuế	114
67/2025/QH15	9	Các khoản chi được trừ và không được trừ khi xác định thu nhập chịu thuế	113
48/2024/QH15	14	Khấu trừ thuế giá trị gia tăng đầu vào	109
22/2021/TT-BGDĐT	14	-	92

SPARQL (`playbook/recipes/r4_qa_articles.rq`):

```
PREFIX vlo:  <https://tmquan.github.io/VLOnto/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
# Articles cited by the most Q&A answers; a clause or point target is
# rolled up to its article with vlo:isPartOf*.
SELECT ?label (COUNT(DISTINCT ?question) AS ?qas) WHERE {
  ?question vlo:hasAnswer ?answer .
  ?answer vlo:answerCitesProvision ?target .
  ?target vlo:isPartOf* ?article .
  ?article a vlo:Article ;
           rdfs:label ?label .
}
GROUP BY ?article ?label
ORDER BY DESC(?qas) ?label
LIMIT 10
```

Output on the full dump (10 rows):

label	qas
Điều 111, 45/2019/QH14	166
Điều 46, 12/2022/NĐ-CP	165
Điều 112, 45/2019/QH14	161
Điều 169, 45/2019/QH14	155
Điều 19, 04/2007/QH12	139
Điều 3, 65/2013/NĐ-CP	130
Điều 44, 38/2019/QH14	114
Điều 9, 67/2025/QH15	113
Điều 14, 48/2024/QH15	109
Điều 14, 22/2021/TT-BGDĐT	92

The most cited article is Article 111 of 45/2019/QH14 (Bộ luật Lao động), *Nghỉ hằng tuần*. The SQL aggregates all 34,869 Q&A article edges; the SPARQL query rolls clause and point targets up to their article with `vlo:isPartOf*`, and both give the same ranking. These are counts over the whole corpus of Q&As; the dashboard shows the same ranking, and the question traces page (section 9) ranks the articles within each legal area.

7.5 Q&As that cite an article

The legal Q&A edges come from the evaluation labels of the vn-legal project. Q&As whose answer cites Article 35 of the Labour Code 45/2019/QH14 (the employee's right to terminate a labour contract):

SQL (`playbook/recipes/r5_qa_citing_article.sql`):

```
-- Legal Q&As whose answer cites Article 35 of the Labour Code 45/2019/QH14
-- (newest first; the text of the Q&A is at the source URL in qas.url).
SELECT q.qa_id,
       q.published_date,
       q.category,
       coalesce(e.clause, '-') AS clause,
       coalesce(e.point, '-') AS point
FROM 'kg/edges_qa_article.parquet' AS e
JOIN 'kg/qas.parquet' AS q USING (qa_id)
WHERE e.dst_article_id = (
    SELECT a.article_id
    FROM 'kg/articles.parquet' AS a
    JOIN 'kg/documents.parquet' AS d USING (doc_id)
    WHERE d.doc_number = '45/2019/QH14'
      AND a.article_no = '35' AND a.segment = 'main')
ORDER BY q.published_date DESC, q.qa_id, clause, point;
```

Output (12 rows):

qa_id	published_date	category	clause	point
138067111	2025-11-11	lao-dong-tien-luong	-	-
138069781	2025-11-10	lao-dong-tien-luong	-	-
138051569	2025-05-27	lao-dong-tien-luong	-	-
138008514	2024-03-14	lao-dong-tien-luong	1	d
138016134	2024-01-01	lao-dong-tien-luong	1	d
138016080	2023-12-23	lao-dong-tien-luong	1	d
138012879	2023-11-27	lao-dong-tien-luong	1	d
138011230	2023-10-05	lao-dong-tien-luong	1	d
138010737	2023-09-25	lao-dong-tien-luong	1	d
138009959	2023-09-09	lao-dong-tien-luong	-	-
138003325	2023-04-18	lao-dong-tien-luong	1	d
138001400	2023-03-14	lao-dong-tien-luong	1	d

SPARQL (`playbook/recipes/r5_qa_citing_article.rq`):

```
PREFIX vlo: <https://tmquan.github.io/VLOnto/ontology#>
# Q&As whose answer cites Article 35 of 45/2019/QH14 or one of its
# clauses or points (newest first)
SELECT DISTINCT ?id ?date WHERE {
  BIND (<https://tmquan.github.io/VLOnto/id/document/333670/dieu-35> AS ?art)
  ?target vlo:isPartOf* ?art .
  ?answer vlo:answerCitesProvision ?target .
  ?question vlo:hasAnswer ?answer ;
            vlo:sourceId ?id .
  OPTIONAL { ?question vlo:publishedDate ?date }
}
ORDER BY DESC(?date) ?id
```

Output on the full dump (first 6 of 12 rows):

id	date
138067111	2025-11-11
138069781	2025-11-10
138051569	2025-05-27
138008514	2024-03-14
138016134	2024-01-01
138016080	2023-12-23

Only identifiers, dates and categories are published for Q&As; open `qas.url` to read a question and its answer at the source.

7.6 Terms of a legal area

Legal terms (*thuật ngữ*) carry their Vietnamese name and an English label whose provenance is recorded (`mt`: machine translation; `site`: given by the source):

SQL (`playbook/recipes/r6_terms_of_area.sql`):

```
-- Legal terms of the area 'Labour and wages', with the provenance of
-- the English label (mt = machine translation, site = from the source).
SELECT t.label_vi,
       coalesce(t.label_en, '-') AS label_en,
       coalesce(t.label_en_source, '-') AS en_source
FROM 'kg/terms.parquet' AS t
JOIN 'kg/areas.parquet' AS a USING (area_id)
WHERE a.label_en = 'Labour and wages'
ORDER BY t.label_vi, t.term_id;
```

Output (first 10 of 246 rows):

label_vi	label_en	en_source
An toàn (trong khai thác mỏ lộ thiên)	Safety (in open-pit mining)	mt
An toàn lao động	Occupational safety	mt
Biên giới khai trường (trong khai thác mỏ lộ thiên)	Mine boundary (in open-pit mining)	mt
Biên giới mỏ lộ thiên	Pit limits	site
Buồng máy (của thang máy)	Machine room	site
Bãi thải (trong khai thác mỏ lộ thiên)	Waste dump	site
Bên nước ngoài tiếp nhận lao động	Foreign receiving party	mt
Bên thuê lại lao động	Labour re-hiring party	mt
Bảo hiểm thất nghiệp	Unemployment insurance	mt
Bệnh nghề nghiệp	Occupational disease	mt

SPARQL (`playbook/recipes/r6_terms_of_area.rq`):

```
PREFIX vlo:  <https://tmquan.github.io/VLOnto/ontology#>
PREFIX skos: <http://www.w3.org/2004/02/skos/core#>
# Legal terms of the area labelled 'Labour and wages' (vi, en)
SELECT ?vi ?en WHERE {
  ?area skos:prefLabel "Labour and wages"@en .
  ?term a vlo:LegalTerm ;
        vlo:legalArea ?area ;
        skos:prefLabel ?vi FILTER (lang(?vi) = "vi")
  OPTIONAL { ?term skos:prefLabel ?en FILTER (lang(?en) = "en") }
}
ORDER BY ?vi ?en
```

Output on the full dump (first 6 of 246 rows):

vi	en
An toàn (trong khai thác mỏ lộ thiên)	Safety (in open-pit mining)
An toàn lao động	Occupational safety
Biên giới khai trường (trong khai thác mỏ lộ thiên)	Mine boundary (in open-pit mining)
Biên giới mỏ lộ thiên	Pit limits
Buồng máy (của thang máy)	Machine room
Bãi thải (trong khai thác mỏ lộ thiên)	Waste dump

7.7 The playground examples

The SPARQL playground of the website offers 10 bilingual example queries, all run on the browser subset (49,846 triples) during the build:

Example	Rows	First result
What replaced the 2012 Labour Code?	1	vlid:document/333670 · 45/2019/QH14 · 2021-01-01 · Bộ luật Lao động số 10/2012/QH13 hết hiệu lực thi hành kể từ ngày Bộ …
Laws in force by area	27	Administrative apparatus · 57
Codes and their validity	30	Bộ luật 20/2000/QH10 · 20/2000/QH10 · unknown
Most amended laws	20	Luật Đầu tư · 61/2020/QH14 · 7
Legal basis of the 2024 Land Law	1	vlid:document/215627 · Hiến pháp Nước cộng hòa xã hội chủ nghĩa Việt Nam năm 2013
Laws that cite the Civil Code 2015	22	vlid:document/621341 · Luật Công nghiệp công nghệ số
Change events by kind	5	amended · 257
Documents issued by the Standing Committee of the National Assembly	2	PL · 145
Superclasses of a Law (ontology)	4	vlo:LegalDocument
Source datasets and licences	4	tmquan/thuvienphapluat-vn-vbpl · 705ade585384108ded5fc30f40a06fe110bdff22 · cc-by-4.0

8. Using VLOnto with the Vietnamese legal assistant agent

8.1 Where VLOnto fits

The companion guide *Building a Vietnamese Legal Assistant Agent with NVIDIA NeMo Helix and Nemotron 3.5* builds an agent that answers questions about Vietnamese law from retrieved articles. Its retrieval service exposes `POST /search`, `GET /article/{doc_number}/{article_no}` and `GET /validity/{doc_number}`, and an MCP server gives the agent three tools on top of it: `search_law`, `get_article` and `check_validity`. A deterministic audit checks every citation of an answer against the service. The guide's measurements show why validity has to come from data: with a stale decree as context, 0 of 30 model answers named the decree that replaced it, and questions about superseded law were flagged 3 of 3 times with the tools against 0 of 3 without them.

The validity data of that service and the change events of VLOnto come from the same validity map (rules v7). VLOnto adds three things the agent can use:

1. **Validity as published, typed data.** Each change event has its kind, effective date, evidence sentence and rule version, and the documents carry the validity class; the parquet files are enough, no GPU service is needed.
2. **The citation network around an article.** What an article cites, which documents cite it, and the legal basis of its act.
3. **Usage signals.** How many published Q&As cite an article, and which other articles they cite together with it.

8.2 Three knowledge-graph tools

`playbook/examples/vlonto_tools.py` implements the three functions over the parquet tables with DuckDB. Each returns a dictionary; `to_text()` renders it as the short text a tool returns to the model.

```
"""Knowledge-graph tools for the Vietnamese legal assistant agent.

Three read-only functions over the VLOnto parquet tables (DuckDB, in process):

  check_validity(doc_number)                  validity, replacing acts and
                                              amendments with their evidence
  expand_citations(doc_number, article_no)    what an article cites, what cites
                                              it, how many Q&A answers cite it
  related_provisions(doc_number, article_no)  articles cited together with
                                              it in Q&A answers

Each function returns a plain dict; to_text() turns it into the short text an
agent tool returns to the model. Co-citation is a usage signal, not a statement
that two provisions are legally related. VLOnto v1 has no court decisions (they
join in v2), so the usage signals come from the published Q&As.
"""
import textwrap

import duckdb

TABLES = ("documents", "articles", "changes", "edges_citation",
          "edges_qa_article")
SUPERSEDING = ("replaced", "repealed")

# co-citation: articles cited in the same Q&A answers as article ?, scored by
# shared * ln(N / df) so that articles cited almost everywhere rank low
RELATED_SQL = """
WITH s AS (SELECT DISTINCT {unit} FROM {table} WHERE dst_article_id = ?),
co AS (SELECT e.dst_article_id AS aid, count(DISTINCT e.{unit}) AS shared
       FROM {table} e JOIN s USING ({unit})
       WHERE e.dst_article_id <> ?
       GROUP BY 1 HAVING count(DISTINCT e.{unit}) >= ?),
df AS (SELECT dst_article_id AS aid, count(DISTINCT {unit}) AS df
       FROM {table} WHERE dst_article_id IN (SELECT aid FROM co) GROUP BY 1),
n AS (SELECT count(DISTINCT {unit}) AS n FROM {table}
```

```
      WHERE dst_article_id IS NOT NULL)
SELECT d.doc_number, a.article_no, a.heading, co.shared, df.df,
       round(co.shared * ln(n.n / df.df), 1) AS score
FROM co JOIN df USING (aid) CROSS JOIN n
JOIN articles a ON a.article_id = co.aid
JOIN documents d ON d.doc_id = a.doc_id
ORDER BY score DESC, d.doc_number, a.article_no
LIMIT ?
"""


CITES_SQL = """
SELECT d.doc_number, t.article_no, sum(e.n_mentions) AS mentions
FROM edges_citation e
JOIN documents d ON d.doc_id = e.dst_doc_id
LEFT JOIN articles t ON t.article_id = e.dst_article_id
WHERE e.src_article_id = ?
GROUP BY ALL
ORDER BY mentions DESC, d.doc_number, t.article_no
LIMIT ?
"""

CITED_BY_SQL = """
SELECT DISTINCT d.doc_number, s.article_no, d.issue_date
FROM edges_citation e
JOIN documents d ON d.doc_id = e.src_doc_id
LEFT JOIN articles s ON s.article_id = e.src_article_id
WHERE e.dst_article_id = ?
ORDER BY d.issue_date DESC NULLS LAST, d.doc_number, s.article_no
"""


class VLOntoTools:
    def __init__(self, kg_dir="kg"):
        self.con = duckdb.connect()
        for t in TABLES:
            path = f"{kg_dir}/{t}.parquet"
            self.con.execute(
                f"CREATE VIEW {t} AS SELECT * FROM read_parquet('{path}')")

    def _rows(self, sql, params=()):
        cur = self.con.execute(sql, list(params))
        cols = [c[0] for c in cur.description]
        return [dict(zip(cols, r)) for r in cur.fetchall()]

    def _count(self, table, unit, aid):
        sql = (f"SELECT count(DISTINCT {unit}) AS n FROM {table} "
               "WHERE dst_article_id = ?")
        return self._rows(sql, [aid])[0]["n"]

    def _article(self, doc_number, article_no):
        rows = self._rows(
            "SELECT a.article_id, a.heading FROM articles a "
            "JOIN documents d USING (doc_id) "
            "WHERE d.doc_number = ? AND a.article_no = ? "
            "AND a.segment = 'main' ORDER BY a.article_id",
            [doc_number, str(article_no)])
        if not rows:
            raise KeyError(f"{doc_number} Điều {article_no}: not in the graph")
        return rows[0]

    def check_validity(self, doc_number):
        docs = self._rows(
            "SELECT doc_id, title_vi, validity, in_force, expiry_date "
            "FROM documents WHERE doc_number = ? ORDER BY doc_id", [doc_number])
        if not docs:
            return {"doc_number": doc_number, "found": False}
        d = docs[0]
        events = self._rows(
            "SELECT c.kind, c.effective_date, a.doc_number AS by_number, "
            "c.evidence FROM changes c "
            "JOIN documents a ON a.doc_id = c.acting_doc_id "
```

```
            "WHERE c.affected_doc_id = ? "
            "ORDER BY c.effective_date NULLS LAST, by_number, c.kind",
            [d["doc_id"]])
        return {
            "doc_number": doc_number, "found": True, "title": d["title_vi"],
            "validity": d["validity"], "in_force": d["in_force"],
            "expiry_date": d["expiry_date"],
            "superseded_by": [e for e in events if e["kind"] in SUPERSEDING],
            "amended_by": [e for e in events if e["kind"] not in SUPERSEDING],
            "same_number_documents": len(docs),
        }

    def expand_citations(self, doc_number, article_no, limit=3):
        art = self._article(doc_number, article_no)
        aid = art["article_id"]
        cited_by = self._rows(CITED_BY_SQL, [aid])
        return {
            "article": f"{doc_number} Điều {article_no}",
            "heading": art["heading"],
            "cites": self._rows(CITES_SQL, [aid, limit]),
            "cited_by_count": len(cited_by),
            "cited_by": cited_by[:limit],
            "qas": self._count("edges_qa_article", "qa_id", aid),
        }

    def related_provisions(self, doc_number, article_no, limit=3,
                           min_shared=2):
        """Articles cited in the same Q&A answers, ranked by
        shared * ln(N / df): the number of shared Q&As weighted by how rare
        the other article is, so that articles cited in almost every answer
        do not dominate."""
        aid = self._article(doc_number, article_no)["article_id"]
        sql = RELATED_SQL.format(table="edges_qa_article", unit="qa_id")
        return {"article": f"{doc_number} Điều {article_no}",
                "in_qas": self._rows(sql, [aid, aid, min_shared, limit])}


def _wrap(text, indent="    ", width=78):
    return textwrap.fill(str(text), width=width, initial_indent=indent,
                         subsequent_indent=indent)


def _art(no):
    return f" Điều {no}" if no else ""


def to_text(result):
    """Render a tool result as the short text a tool returns to the model."""
    lines = []
    if "validity" in result:
        lines.append(f"{result['doc_number']} ({result['title']}): "
                     f"validity {result['validity']}, "
                     f"in force: {result['in_force']}")
        for e in result["superseded_by"]:
            lines.append(f"  {e['kind']} by {e['by_number']} "
                         f"from {e['effective_date']}")
            lines.append(_wrap(f"evidence: {e['evidence']}"))
        for e in result["amended_by"][-3:]:
            lines.append(f"  {e['kind']} by {e['by_number']} "
                         f"from {e['effective_date']}")
        hidden = len(result["amended_by"]) - 3
        if hidden > 0:
            lines.append(f"  ({hidden} earlier amendments not shown)")
    elif "cites" in result:
        lines.append(f"{result['article']}: {result['heading']}")
        for c in result["cites"]:
            lines.append(f"  cites {c['doc_number']}{_art(c['article_no'])}")
        lines.append(f"  cited by {result['cited_by_count']} documents or "
                     f"articles, newest first:")
        for c in result["cited_by"]:
            lines.append(f"    {c['doc_number']}{_art(c['article_no'])} "
                         f"(issued {c['issue_date']})")
```

```
        lines.append(f"  cited in {result['qas']:,} Q&A answers")
    else:
        lines.append(f"{result['article']}: provisions cited together with it")
        lines.append("  in Q&As (shared / all citing it / score):")
        for r in result["in_qas"]:
            head = textwrap.shorten(r["heading"] or "-", width=52,
                                    placeholder=" …")
            lines.append(f"    {r['doc_number']}{_art(r['article_no'])}: "
                         f"{head}")
            lines.append(f"      {r['shared']:,} / {r['df']:,} / "
                         f"{r['score']}")
    return "\n".join(lines)
```

Calling the tools as an agent would (`playbook/examples/agent_demo.py`):

```
"""Call the three VLOnto tools the way an agent tool server would."""
from vlonto_tools import VLOntoTools, to_text

kg = VLOntoTools("kg")
print(to_text(kg.check_validity("10/2012/QH13")))
print()
print(to_text(kg.expand_citations("45/2019/QH14", "35")))
print()
print(to_text(kg.related_provisions("45/2019/QH14", "169")))
```

Output:

```
10/2012/QH13 (Bộ luật Lao động): validity expired, in force: False
  replaced by 45/2019/QH14 from 2021-01-01
    evidence: Bộ luật Lao động số 10/2012/QH13 hết hiệu lực thi hành kể từ
    ngày Bộ luật này có hiệu lực.
  amended by 92/2015/QH13 from 2016-07-01
  partially_repealed by 92/2015/QH13 from 2016-07-01

45/2019/QH14 Điều 35: Quyền đơn phương chấm dứt hợp đồng lao động của người lao động
  cited by 2 documents or articles, newest first:
    374/2025/NĐ-CP Điều 25 (issued 2025-12-31)
    22/2022/TT-BCA (issued 2022-05-10)
  cited in 12 Q&A answers

45/2019/QH14 Điều 169: provisions cited together with it
  in Q&As (shared / all citing it / score):
    135/2020/NĐ-CP Điều 4: Tuổi nghỉ hưu trong điều kiện lao động bình thường
      20 / 20 / 140.5
    58/2014/QH13 Điều 54: Điều kiện hưởng lương hưu
      5 / 22 / 34.7
    41/2024/QH15 Điều 2: Đối tượng tham gia bảo hiểm xã hội bắt buộc và bảo …
      5 / 29 / 33.3
```

The first result is what the agent needs to answer a question asked under the 2012 code: the act is no longer in force, the replacing act and its effective date, and the sentence that says so. The second shows how widely an article is used. The third ranks articles by the number of Q&As that cite both, weighted by how rare the other article is; it is a usage signal, not a legal relation, and Q&A article links are evaluation labels whose spot check found 9 of 10 correct.

8.3 Plugging the tools into the agent

The following integration is a design that follows the guide's architecture; it was not run against the live agent for this playbook.

- **As MCP tools.** Register the three functions next to `search_law`, `get_article` and `check_validity` in the agent's MCP server, each returning `to_text(...)`. Install the server as a console script in the NeMo Helix environment, as the guide does for its own tools (stdio MCP servers start in the agent's workspace, so relative script paths fail). A minimal server with the `mcp` Python SDK:

```
from mcp.server.fastmcp import FastMCP
from vlonto_tools import VLOntoTools, to_text
```

```
mcp = FastMCP("vlonto-kg")
kg = VLOntoTools("/data/vlonto/build/kg")    # kg/ folder of a local build


@mcp.tool()
def kg_check_validity(doc_number: str) -> str:
    """Validity, replacing acts and amendments of a Vietnamese legal document."""
    return to_text(kg.check_validity(doc_number))


@mcp.tool()
def kg_expand_citations(doc_number: str, article_no: str) -> str:
    """What an article cites, what cites it, and how many Q&As cite it."""
    return to_text(kg.expand_citations(doc_number, article_no))


@mcp.tool()
def kg_related_provisions(doc_number: str, article_no: str) -> str:
    """Articles cited together with this one in Q&A answers."""
    return to_text(kg.related_provisions(doc_number, article_no))


if __name__ == "__main__":
    mcp.run()
```

- **In the retrieval service.** `GET /validity/{doc_number}` and `check_validity` read the same relations, so a deployment without the GPU-backed service can answer validity questions from the parquet files alone. Keep one source of truth per release: rebuild VLOnto whenever the validity map is regenerated (section 11).
- **In the answer audit.** The audit parses each citation (`[45/2019/QH14, Điều 35]`) and checks that the article exists and that its act is in force. With VLOnto it can also append the amendment history of the act and flag citations of an act whose validity is unknown instead of treating them as current.
- **In the prompt.** Keep the guide's rule: when a tool reports validity `unknown`, the agent says that validity is not confirmed rather than implying that the provision is current.

8.4 Cautions

- Validity is unknown for 70.2% of the documents; a missing change event does not mean that an act is in force.
- The change events are extracted from text with a precision of 29/30 on a sample (section 6.5); show the evidence sentence and the source URL with every validity claim.
- Citation and usage counts reflect the corpus (its coverage of documents and Q&As), not legal importance.

9. The website and the 3D playgrounds

The website is a set of static pages generated from the build's data bundles. It is published at https://tmquan.github.io/VLOnto/, which opens the 3D ontology playground, and it also opens from disk (section 2.1).

Page	What it shows	Data
`index.html`	The start page: the 3D ontology playground, the ontology as a 3D graph: 75 nodes (classes, concept schemes and the external terms they are aligned to) and 104 links (sub-class 24, property 40, scheme membership 13, alignment 27), coloured by module.	`ontology_graph.json`
`about.html`	Overview of VLOnto with the key counts of the build and links to every view.	`stats.json`
`ontology-3d.html`	Only a redirect to `index.html`, so that older links to the playground keep working.	-
`network-3d.html`	The legal network: 40,816 documents in 3D with the legal-force rank as height, and their change (15,035), legal-basis (99,038) and citation (152,251 document pairs) links. Phones load a lighter bundle of 6,000 documents.	`legal_network.json`, `legal_network_lite.json`
`sparql.html`	A SPARQL editor that runs Oxigraph, compiled to WebAssembly, inside the page over the browser subset (49,846 triples), with the 10 examples of section 7.7.	`vlonto_core_ttl.js`, `sparql_examples.json`
`dashboard.html`	Charts of documents by type, year, issuer, area and validity; change events by kind and year; legal-basis methods and citation kinds; the most cited documents and articles; Q&As, terms and news.	`dashboard.json`
`traces.html`	Question traces: for each of the 27 legal areas of the Q&As, the 20 articles that the most answers in the area cite, each traced from one question and its answer through the cited point, clause and article to the chapter and the law, highlighted like Figure 6 of the paper (540 traces). It shows question titles, never answer text.	`data/traces/`
`ontology.html`	The ontology documentation generated by pyLODE from the Turtle file, with every class, property and concept scheme.	`vlonto.html`
`playbook.html`	This playbook as HTML, with the PDF and the Markdown source as downloads.	`playbook_html.js`
`downloads.html`	The files that need no build (section 2.2): the ontology, the core Turtle subset, the data dictionary, the website's JSON data, the licence policy and attribution, the figures and this playbook, with a checksum list.	`downloads/`, `SHA256SUMS.txt`

Offline use. Every page is plain HTML with classic scripts: no ES modules, no `fetch`, no server. Data bundles are JavaScript files that assign the JSON to `window.VLONTO` (generated by `tools/json2js.py`), the WebAssembly of Oxigraph is embedded in its script, and the fonts are in the site folder, so the pages work from `file://` without network access. Each bundle stays below 20 MB (15 files listed in `manifest.json` with size and sha256).

3D views. The 3D views use three.js and 3d-force-graph with text labels drawn as sprites. The legal-network positions are precomputed in the build (`vlonto.cli layout`), so the page does not run a force simulation on 40,816 nodes. Rendering speed depends on the graphics hardware; the automated website checks render with a software WebGL implementation, so their frame rates say nothing about a real GPU.

10. Reproducibility

10.1 Pinned tools and libraries

Component	Version	Used for
Typst	0.15.1 (9dfd3a08)	PDF of this playbook
pandoc	3.9	Markdown to Typst and HTML
Python (pinned toolchain)	3.12.16	website build and checks (tools/setup_tools.sh)
Python (reference build)	3.11.18	knowledge-graph build and playbook tools of VLOnto v1 (macOS, arm64)
uv	0.13.7	Python environments
Node.js / npm	24.19.0 / 11.17.0	vendored JavaScript build
esbuild	0.28.2	browser bundles
Chrome headless shell	157.0.8103.50	website checks
three.js	0.187.0	3D rendering
3d-force-graph	1.80.2	3D graphs
three-spritetext	1.10.0	3D labels
ECharts	6.1.0	dashboard charts
Oxigraph (npm)	0.5.13	SPARQL in the browser
JetBrains Mono	2.304	code font (NL variant)
duckdb	2.0.0	knowledge-graph build
igraph	1.0.0	knowledge-graph build
owlrl	7.6.2	knowledge-graph build
polars	2.1.0	knowledge-graph build
pyarrow	26.0.1	knowledge-graph build
pylode	3.6.1	knowledge-graph build
pyoxigraph	0.5.13	knowledge-graph build
pyshacl	0.40.2	knowledge-graph build
rdflib	7.6.0	knowledge-graph build

The fonts come from fixed commits of their repositories (Be Vietnam Pro, Noto Serif, Noto Sans Math, Noto Sans Symbols 2) or a fixed release (JetBrains Mono), and only these files are used for the PDF (16 font files in `typst/fonts`).

10.2 Lock files

- `tools/chrome-deps.conda.lock`: conda packages of the libraries Chrome needs on the node.
- `tools/requirements-web.lock`: hash-checked Python packages of the web and PDF environment.
- `tools/vendor.lock.json`: package, version, file, sha256 and licence of every vendored website file (98 files).
- `tools/versions.lock`: versions, download URLs and sha256 of the downloaded tools and fonts.

`tools/setup_tools.sh --check` verifies every pinned tool and exits with an error on any drift; `tools/fetch_vendor.sh --check-repo` compares the vendored files with `vendor.lock.json`.

10.3 Deterministic build of this playbook

The Markdown is generated, never edited by hand: `playbook.md.j2` + `data/facts.json` → `playbook.md` → pandoc → `playbook.typ` → Typst → PDF. `scripts/build_pdf.sh` runs the chain:

```
bash scripts/build_pdf.sh --facts       # 1. gather facts (needs the KG build)
bash scripts/build_pdf.sh               # 2. Markdown, SVG figures, Typst, PDF, HTML
bash scripts/build_pdf.sh --check       # 3. check the PDF, rebuild it twice
bash scripts/build_pdf.sh --check-files # 4. printed sizes and sha256 vs the files
bash scripts/build_pdf.sh --pdf-only    # Typst only, from the generated playbook.typ
```

`playbook/Makefile` offers the same steps as targets: `make -C playbook facts`, `pdf`, `check`, `check-files` and `pdf-only`; `make -C playbook` runs `pdf`, `check` and `check-files`. `--facts` loads the RDF dump into a

new Oxigraph store for the recipes; `REUSE_STORE=1` keeps the previous store and is only for runs on an unchanged dump.

Typst runs with `--font-path typst/fonts --ignore-system-fonts --ignore-embedded-fonts`, so only the vendored fonts can be used, and with a fixed `SOURCE_DATE_EPOCH` (the build date), so two builds of the same sources are byte-identical. `--check` runs `tools/pdf_check.py`, which checks that every font is embedded and vendored, that no glyph is missing, that text stays inside the margins, that no lines overlap and that no page is blank, and rebuilds the PDF twice to compare the bytes; it then runs `playbook/tools/text_collisions.py`, which finds glyphs that run into other text on the same baseline, such as a word wider than its table cell. The generated `playbook.typ`, the figure data and the SVG figures are kept in the repository, so the PDF can be rebuilt with Typst alone: the GitHub Pages workflow (job `reproduce-pdf`) rebuilds it on Linux with the pinned Typst release (`tools/fetch_typst.sh`, sha256-checked) and compares it byte for byte with `site/downloads/VLOnto-playbook.pdf`, and the preview workflow runs the same job on preview branches. The figures are drawn by Typst from `data/figures.json` (`charts.typ`): natively in the PDF and as SVG for the Markdown and the website.

10.4 Checksums of the reference build

`playbook/data/SHA256SUMS` records the sha256 of the outputs of the reference build of 2026-11-12 that this playbook describes: the tables and RDF dumps (paths relative to the build folder) and the ontology files (relative to the repository). The first 16 hexadecimal digits are:

File	Size	sha256 (prefix)
kg/areas.parquet	7 kB	d2991b4c9328458f
kg/articles.parquet	9.3 MB	1b4f587a82a88dc6
kg/changes.parquet	1.3 MB	8ac8c354d628c083
kg/data_dictionary.json	29 kB	a1cebf64f35c5dbd
kg/data_dictionary.md	15 kB	1694da8bdfdd485d
kg/doc_types.parquet	3 kB	4492320fcc76b562
kg/documents.parquet	4.9 MB	9fa21cf00c273965
kg/edges_change.parquet	230 kB	6c020502418fe78e
kg/edges_citation.parquet	3.1 MB	680a38acc145bbd4
kg/edges_legal_basis.parquet	667 kB	f243d3f950ca9e75
kg/edges_news_document.parquet	12 kB	e8fc672c8674e61d
kg/edges_qa_article.parquet	461 kB	12afaab65c4a4640
kg/edges_qa_document.parquet	806 kB	8844ff35b0e289f3
kg/edges_term_related.parquet	75 kB	b8d838592da46701
kg/issuers.parquet	7 kB	93504e8c6f7326c5
kg/legal_force_ranks.parquet	3 kB	ffe962e40c5851c9
kg/news.parquet	31 kB	6696aed7dc3fe639
kg/news_categories.parquet	3 kB	0fa3a472c5b03aa1
kg/provisions_ref.parquet	1.0 MB	4f310ecb270f06de
kg/qas.parquet	3.9 MB	c412cf700967d0e4
kg/sources.parquet	5 kB	04bd4d6ee590815c
kg/structure.parquet	1.6 MB	392381bc225cfbf8
kg/terms.parquet	481 kB	9a4a119e87567e5e
ontology/vlonto-context.jsonld	7 kB	a7d3f1b413e398a9
ontology/vlonto-shapes.ttl	13 kB	8aec3dc3d0462985
ontology/vlonto.html	680 kB	b57be5456912b0c5
ontology/vlonto.ttl	122 kB	5c49e90eac33eb0d
rdf/vlonto-core.ttl	2.6 MB	b7c503b0dc5cbd36
rdf/vlonto-kg.nt.gz	48.2 MB	781db886ced8468d

The ontology files, `rdf/vlonto-core.ttl` and the data dictionary on the website's downloads page are these files; check a download with `sha256sum` against this list. `bash scripts/build_pdf.sh --check-files` (`playbook/tools/check_files.py`) compares every size and sha256 printed in this playbook, and the triple counts of section 5.3, with the files of the build folder, the repository and the downloads page.

A local rebuild is checked by its counts (section 6.2) rather than by these bytes: byte equality of rebuilt tables and dumps is not tested.

11. Maintenance runbook

11.1 New revisions of the source datasets

1. Update the revision of each changed dataset in `vlonto/vlonto/config.py` (`SOURCES`) and in the vn-legal project, and download the new snapshots.
2. Re-run the vn-legal data pipeline: clean and scope, split into articles, extract the validity map, build the evaluation labels.
3. Run `python -m vlonto.cli all` (section 5) and read `build/reports/kg_report.md`: all checks must pass and the reconciliation items must match.
4. Compare `build/site_data/stats.json` with the previous release and explain every large change.
5. Draw new spot-check samples for the edge types whose extraction rules or inputs changed, judge them and record the verdicts in `docs/spot_check_review.json`; then run `python -m vlonto.cli report`.
6. Re-check the licences of the dataset cards (`docs/licence_policy.md`) if a card changed.
7. Rebuild the website data and this playbook (`bash scripts/build_pdf.sh --facts` without `REUSE_STORE`, `bash scripts/build_pdf.sh`, `bash scripts/build_pdf.sh --check` and, once the website's downloads folder holds the new files, `bash scripts/build_pdf.sh --check-files`), and run `tools/site_check.py` on the website.

11.2 Re-running the validity rules

The validity map belongs to the vn-legal project (`$R/data/validity/relations.parquet` and `$R/data/validity/validity.parquet`). To change the rules or the as-of date:

1. Edit and re-run the rules there (`SAMPLE_SEED=31 $R/envs/data/bin/python $R/code/validity.py`, as in the guide) and keep the known-case checks passing.
2. Judge a fresh sample of relations and record the precision with the rule version.
3. Set `VALIDITY_AS_OF` in `vlonto/vlonto/config.py` and rebuild VLOnto: every change event records the rule version, and the build activity records the as-of date.
4. Rebuild the retrieval service of the agent from the same map, so that the agent and VLOnto agree.

11.3 Releasing

1. Set the version (`__version__`) and the release name (`RELEASE`, for example `v1`) in `vlonto/vlonto/__init__.py` and `BUILD_DATE` in `vlonto/vlonto/config.py` (or `VLONTO_BUILD_DATE`); update `owl:versionInfo`, `owl:versionIRI`, `owl:priorVersion` and `dcterms:modified` in `ontology/vlonto.ttl` following the policy of section 4.6.
2. Rebuild the graph, the website data and the playbook; all checks must pass.
3. Regenerate `playbook/data/SHA256SUMS` (`bash scripts/build_pdf.sh --facts`), refresh the website's downloads folder and its `SHA256SUMS.txt` (the GitHub Pages workflow checks that list with `sha256sum -c`), run `bash scripts/build_pdf.sh --check-files`, push the repository to https://github.com/tmquan/VLOnto so that GitHub Pages publishes the website at https://tmquan.github.io/VLOnto/, and tag the repository with the release name (`v1`). No data is uploaded: the source datasets stay the data home, and users build the knowledge graph locally.
4. Keep the ontology file of the previous version available; its version IRI must keep resolving.

12. Limitations and roadmap

12.1 Limitations of the knowledge graph

From the build report of 2026-11-12:

- Citation resolution to an in-scope document: external citations 83.1%, external mentions 68.7%, parent law 97.0%, parent-law mentions 99.0%; the rest cite documents outside the corpus or are too vague. When the citing document has no date, a law cited by name resolves to the only or latest version, which can be an older law of the same name (257/1998/TT-BGTVT linked to the 1987 Foreign Investment Law).
- Legal basis: errors in the source parent-acts metadata are carried over (1 of 10 sampled), and a document mentioned in a preamble can be a cross-reference rather than a legal basis (1 of 10 not decidable).
- Document issue dates: 144 source dates more than one year away from the year in the document number were set to null (kept in issue_date_source); 170 documents have no issue date.
- Validity: 28,660 of 40,816 documents have validity 'unknown' (validity pipeline as of 2026-11-08).
- Issuers: 57 curated issuing bodies; 1,750 documents have no issuer and 1,204 only symbols without a curated code; joint issuers without a curated code are not linked.
- News: the source gives a publication date for only a small part of the news items (87 of 681).
- Glossary: English labels of terms are mostly machine-translated in the source; definitions are not republished (licence).
- Texts: the knowledge-graph tables and dumps hold no article or document bodies (article texts stay in the build's local-only article-text table), no question, answer or news text, no glossary definitions and no person-name fields (signers, authors, editors); they keep document titles, headings of at most 200 characters and change-evidence quotes of at most 300 characters. The website's Traces page shows the titles of the questions it traces (licence policy, section 2.2). Titles and article headings of public legal documents can name officials or award recipients (a pattern scan finds such names in 206 article headings and 6 document titles); they are kept as published by the source.
- Spot checks are small samples (10 per edge type); they catch systematic errors but give only coarse precision estimates.

12.2 Limitations of the website and this playbook

- The website is machine-checked only in Chromium (headless Chrome); Safari and Firefox behaviour under `file://` follows the same rules but must be checked by hand.
- The 3D views were tested with software WebGL; large graphs can be slow on weak graphics hardware, which is why phones get the lighter network bundle.
- The playbook's numbers are as good as the build reports; a fact that the reports do not contain is not stated.
- The agent integration of section 8.3 is a design; only the knowledge-graph tools themselves were run.

12.3 Roadmap

1. Close the validity gap: add official validity sources and consolidated texts as evidence, and measure recall of the change events.
2. Measure precision with larger, stratified samples and with legal review, starting with Q&A article labels and legal-basis links.
3. VLOnto v2: add the court datasets `anle-toaan-gov-vn` and `cbba-toaan-gov-vn` (court decisions, precedents and the provisions they apply).
4. Resolve joint issuers that have no curated code.
5. Publish the agent tools as an MCP server package and evaluate them in the agent's test set.
6. Serve the RDF with content negotiation and a public SPARQL endpoint.

References

- Khac-Hoai Nam Bui and Quan Tran Minh. *Vietnamese Legal Ontology: Every legal answer, traced to the article that grounds it.* arXiv:2611.11426 [cs.CL], 2026. https://arxiv.org/abs/2611.11426

Appendix A. Namespaces and prefixes

Prefix	Namespace
vlo:	https://tmquan.github.io/VLOnto/ontology#
vlid:	https://tmquan.github.io/VLOnto/id/
eli:	http://data.europa.eu/eli/ontology#
skos:	http://www.w3.org/2004/02/skos/core#
dcterms:	http://purl.org/dc/terms/
prov:	http://www.w3.org/ns/prov#
schema:	https://schema.org/
org:	http://www.w3.org/ns/org#
owl:	http://www.w3.org/2002/07/owl#
rdfs:	http://www.w3.org/2000/01/rdf-schema#
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#
xsd:	http://www.w3.org/2001/XMLSchema#
vann:	http://purl.org/vocab/vann/

Appendix B. Ontology reference

Generated from `ontology/vlonto.ttl` (version 1.0.0, 2,061 triples): every class, property and concept scheme of VLOnto by module, with its English and Vietnamese labels, its axioms, its English definition and, where it has one, its English scope note.

B.1 Module documents (*văn bản*)

Legal documents and their types.

Classes

- **`vlo:Circular`** · Circular · *Thông tư*. Subclass of `vlo:NormativeLegalDocument`. A normative document issued by a minister or the head of a ministerial-level agency, the Chief Justice of the Supreme People's Court, the Prosecutor General or the State Auditor General.
- **`vlo:Code`** · Code · *Bộ luật*. Subclass of `vlo:NormativeLegalDocument`. A comprehensive law adopted by the National Assembly that systematically regulates a whole field, such as the Civil Code or the Labour Code.
- **`vlo:Constitution`** · Constitution · *Hiến pháp*. Subclass of `vlo:NormativeLegalDocument`. The fundamental law of the State, adopted by the National Assembly; it has the highest legal force.
- **`vlo:Decision`** · Decision · *Quyết định*. Subclass of `vlo:NormativeLegalDocument`. A normative decision issued by the Prime Minister, a minister or another competent head of a central body; only decisions numbered in the normative form N/YYYY/CODE are in scope.
- **`vlo:Decree`** · Decree · *Nghị định*. Subclass of `vlo:NormativeLegalDocument`. A normative document issued by the Government, usually detailing or implementing laws.
- **`vlo:JointCircular`** · Joint circular · *Thông tư liên tịch*. Subclass of `vlo:NormativeLegalDocument`. A circular issued jointly by two or more ministers, heads of ministerial-level agencies or judicial bodies.
- **`vlo:JointResolution`** · Joint resolution · *Nghị quyết liên tịch*. Subclass of `vlo:NormativeLegalDocument`. A resolution issued jointly by a state body and the central body of a socio-political organisation, for example the Standing Committee of the National Assembly or the Government together with the Presidium of the Viet Nam Fatherland Front.
- **`vlo:Law`** · Law · *Luật*. Subclass of `vlo:NormativeLegalDocument`. A statute adopted by the National Assembly.
- **`vlo:LegalDocument`** · Legal document · *Văn bản pháp luật*. Subclass of `eli:LegalResource`, `schema:Legislation`; disjoint with `vlo:Provision`. A document issued by a competent state body that has legal effect, identified by its type, number and issue date.
- **`vlo:NormativeLegalDocument`** · Normative legal document · *Văn bản quy phạm pháp luật*. Subclass of `vlo:LegalDocument`. A legal document containing generally binding rules of conduct, issued in the form, order and procedure set out by the Law on Promulgation of Legal Normative Documents.
- **`vlo:Ordinance`** · Ordinance · *Pháp lệnh*. Subclass of `vlo:NormativeLegalDocument`. A normative document adopted by the Standing Committee of the National Assembly on matters assigned by the National Assembly.
- **`vlo:Resolution`** · Resolution · *Nghị quyết*. Subclass of `vlo:NormativeLegalDocument`. A normative resolution issued by a collegial body such as the National Assembly, its Standing Committee, the Government or the Council of Judges of the Supreme People's Court.

Object properties

- **`vlo:documentType`** · document type · *loại văn bản*. Sub-property of `eli:type_document`; domain `vlo:LegalDocument`; range `vlo:DocumentType`; `owl:FunctionalProperty`. The type of a legal document as a concept of the DocumentType scheme.

Datatype properties

- **`vlo:documentNumber`** · document number · *số, ký hiệu văn bản*. Sub-property of `eli:id_local`; domain `vlo:LegalDocument`; range `xsd:string`. The official number and symbol of a legal document as printed in its header, for example 45/2019/QH14.

- **`vlo:documentNumberKey`** · document number key · *khóa số văn bản*. domain `vlo:LegalDocument`; range `xsd:string`. A normalised lookup key of the document number (ASCII, upper case, unified separators, Đ written as D), used to match citations to documents.
- **`vlo:issueDate`** · issue date · *ngày ban hành*. Sub-property of `eli:date_document`; domain `vlo:LegalDocument`; range `xsd:date`. The date on which a legal document was adopted or signed.
- **`vlo:title`** · title · *tên văn bản*. Sub-property of `dcterms:title`; range `rdfs:Literal`. The Vietnamese title of a legal document, including its type and, for sub-statutory acts, its number.

B.2 Module provisions (*điều khoản*)

Structural units of legal documents: parts, chapters, sections, articles, clauses, points and appendices.

Classes

- **`vlo:Appendix`** · Appendix · *Phụ lục*. Subclass of `vlo:Provision`. An attachment issued with a legal document, such as a regulation, charter or list promulgated by the document; it may contain its own articles.
- **`vlo:Article`** · Article · *Điều*. Subclass of `vlo:Provision`. The basic numbered unit of a legal document (Điều N), usually with a heading, divided into clauses.
- **`vlo:Chapter`** · Chapter · *Chương*. Subclass of `vlo:Provision`. A division of a legal document grouping sections or articles under a heading (Chương I, Chương II, …).
- **`vlo:Clause`** · Clause · *Khoản*. Subclass of `vlo:Provision`. A numbered subdivision of an article (khoản 1, khoản 2, …).
- **`vlo:Part`** · Part · *Phần*. Subclass of `vlo:Provision`. A top-level division of a long legal document, grouping chapters (Phần thứ nhất, Phần thứ hai, …).
- **`vlo:Point`** · Point · *Điểm*. Subclass of `vlo:Provision`. A lettered subdivision of a clause (điểm a, điểm b, …).
- **`vlo:Provision`** · Provision · *Đơn vị cấu trúc của văn bản*. Subclass of `eli:LegalResourceSubdivision`. A structural unit of a legal document, such as a part, chapter, section, article, clause, point or appendix.
- **`vlo:Section`** · Section · *Mục*. Subclass of `vlo:Provision`. A division of a chapter grouping articles (Mục 1, Mục 2, …).

Object properties

- **`vlo:hasPart`** · has part · *gồm*. Sub-property of `eli:has_part`; domain `eli:LegalResource`; range `vlo:Provision`. Inverse of vlo:isPartOf: links a document or provision to a provision it immediately contains. Not materialised in the RDF dumps.
- **`vlo:inDocument`** · in document · *thuộc văn bản*. Sub-property of `eli:is_part_of`; domain `vlo:Provision`; range `vlo:LegalDocument`; `owl:FunctionalProperty`. Links a provision at any depth directly to the legal document that contains it.
- **`vlo:isPartOf`** · is part of · *thuộc*. Sub-property of `eli:is_part_of`; domain `vlo:Provision`; range `eli:LegalResource`; inverse of `vlo:hasPart`. Links a provision to the unit that immediately contains it: a section, chapter, part, appendix, article, clause or the document itself.

Datatype properties

- **`vlo:heading`** · heading · *tên điều khoản*. Sub-property of `dcterms:title`; domain `vlo:Provision`; range `rdfs:Literal`. The heading of a provision as printed, cut at 200 characters (for example the article title or the chapter heading).
- **`vlo:position`** · position · *thứ tự*. domain `vlo:Provision`; range `xsd:integer`. The 1-based order of a provision among the provisions of the same kind in its document, following the text.
- **`vlo:provisionNumber`** · provision number · *số hiệu điều khoản*. Sub-property of `dcterms:identifier`; domain `vlo:Provision`; range `xsd:string`. The number or letter of a provision within its parent: 35 or 5a for an article, 2 for a clause, a or đ for a point, III for a chapter, 1 for an appendix.

B.3 Module agents (*chủ thể*)

Public authorities that issue documents or decide cases.

Classes

- **`vlo:Court`** · Court · *Tòa án*. Subclass of `vlo:PublicAuthority`. A people's court or military court that decides cases; identified by its official name, for example Tòa án nhân dân tỉnh Cà Mau. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:IssuingBody`** · Issuing body · *Cơ quan ban hành*. Subclass of `vlo:PublicAuthority`. A public authority that issues legal documents, identified in the knowledge graph by the issuer code used in document numbers (for example BTC for the Ministry of Finance).
- **`vlo:PublicAuthority`** · Public authority · *Cơ quan nhà nước*. Subclass of `org:Organization`. A state organisation or office holder acting for the State, such as the National Assembly, the Government, a ministry, the Prime Minister or a court.

Object properties

- **`vlo:issuedBy`** · issued by · *cơ quan ban hành*. Sub-property of `eli:passed_by`; domain `vlo:LegalDocument`; range `vlo:IssuingBody`. A body that issued a legal document; joint documents have several issuing bodies.

Datatype properties

- **`vlo:issuerCode`** · issuer code · *ký hiệu cơ quan ban hành*. domain `vlo:IssuingBody`; range `xsd:string`. The abbreviation that identifies an issuing body in document numbers, written in ASCII upper case (for example BTC, BLDTBXH, TTG).

B.4 Module validity and change (*hiệu lực và thay đổi*)

Validity status of documents and the dated events that replace, amend, repeal or terminate them.

Classes

- **`vlo:ChangeKind`** · Change kind · *Loại thay đổi*. Subclass of `skos:Concept`. A kind of legal change: replaced, amended, repealed, partially repealed or partially terminated.
- **`vlo:LegalChange`** · Legal change · *Sự kiện thay đổi hiệu lực văn bản*. Subclass of `prov:Activity`. A dated event in which an acting legal document replaces, amends, repeals, partially repeals or partially terminates an affected legal document. The event is reified so that its kind, effective date, the evidence sentence and the extraction method can be recorded; the direct shortcut properties (vlo:replaces, vlo:amends, …) carry the same information without the details.
- **`vlo:ValidityStatus`** · Validity status · *Tình trạng hiệu lực*. Subclass of `eli:InForce`, `skos:Concept`. The validity of a legal document as of the knowledge-graph date, combining the curated source status with text-extracted change relations (validity rules v7). It refines the ELI in-force concept with eight values.

Object properties

- **`vlo:actingDocument`** · acting document · *văn bản tác động*. domain `vlo:LegalChange`; range `vlo:LegalDocument`; `owl:FunctionalProperty`. The legal document whose text makes the change (the newer act).
- **`vlo:affectedDocument`** · affected document · *văn bản bị tác động*. domain `vlo:LegalChange`; range `vlo:LegalDocument`; `owl:FunctionalProperty`. The legal document that is replaced, amended, repealed or terminated by the change.
- **`vlo:amendedBy`** · amended by · *được sửa đổi, bổ sung bởi*. domain `vlo:LegalDocument`; range `vlo:LegalDocument`. Inverse of vlo:amends. Not materialised in the RDF dumps.
- **`vlo:amends`** · amends · *sửa đổi, bổ sung*. Sub-property of `eli:amends`, `vlo:changes`; domain `vlo:LegalDocument`; range `vlo:LegalDocument`; inverse of `vlo:amendedBy`. The subject document amends or supplements some provisions of the object document (sửa đổi, bổ sung).
- **`vlo:changeKind`** · change kind · *loại thay đổi*. domain `vlo:LegalChange`; range `vlo:ChangeKind`; `owl:FunctionalProperty`. The kind of a legal change.
- **`vlo:changedBy`** · changed by · *bị thay đổi bởi*. domain `vlo:LegalDocument`; range `vlo:LegalDocument`. Inverse of vlo:changes. Not materialised in the RDF dumps.

- **`vlo:changes`** · changes · *thay đổi*. Sub-property of `eli:changes`; domain `vlo:LegalDocument`; range `vlo:LegalDocument`; inverse of `vlo:changedBy`. Shortcut for any legal change: the subject document replaces, amends, repeals, partially repeals or partially terminates the object document.
- **`vlo:partiallyRepealedBy`** · partially repealed by · *bị bãi bỏ một phần bởi*. domain `vlo:LegalDocument`; range `vlo:LegalDocument`. Inverse of vlo:partiallyRepeals. Not materialised in the RDF dumps.
- **`vlo:partiallyRepeals`** · partially repeals · *bãi bỏ một phần*. Sub-property of `vlo:changes`; domain `vlo:LegalDocument`; range `vlo:LegalDocument`; inverse of `vlo:partiallyRepealedBy`. The subject document repeals some articles, clauses, points or contents of the object document, which otherwise stays in force.
- **`vlo:partiallyTerminatedBy`** · partially terminated by · *bị làm hết hiệu lực một phần bởi*. domain `vlo:LegalDocument`; range `vlo:LegalDocument`. Inverse of vlo:partiallyTerminates. Not materialised in the RDF dumps.
- **`vlo:partiallyTerminates`** · partially terminates · *làm hết hiệu lực một phần*. Sub-property of `vlo:changes`; domain `vlo:LegalDocument`; range `vlo:LegalDocument`; inverse of `vlo:partiallyTerminatedBy`. The subject document declares that some provisions of the object document cease to be in force (hết hiệu lực một phần).
- **`vlo:repealedBy`** · repealed by · *bị bãi bỏ bởi*. domain `vlo:LegalDocument`; range `vlo:LegalDocument`. Inverse of vlo:repeals. Not materialised in the RDF dumps.
- **`vlo:repeals`** · repeals · *bãi bỏ*. Sub-property of `eli:repeals`, `vlo:changes`; domain `vlo:LegalDocument`; range `vlo:LegalDocument`; inverse of `vlo:repealedBy`. The subject document repeals the object document as a whole without replacing it (bãi bỏ).
- **`vlo:replacedBy`** · replaced by · *được thay thế bởi*. domain `vlo:LegalDocument`; range `vlo:LegalDocument`. Inverse of vlo:replaces. Not materialised in the RDF dumps.
- **`vlo:replaces`** · replaces · *thay thế*. Sub-property of `eli:repeals`, `vlo:changes`; domain `vlo:LegalDocument`; range `vlo:LegalDocument`; inverse of `vlo:replacedBy`. The subject document replaces the object document as a whole, which then ceases to be in force (thay thế).
- **`vlo:validityStatus`** · validity status · *tình trạng hiệu lực*. Sub-property of `eli:in_force`; domain `vlo:LegalDocument`; range `vlo:ValidityStatus`; `owl:FunctionalProperty`. The validity of a legal document as of the build date of the knowledge graph.

Datatype properties

- **`vlo:changeEffectiveDate`** · change effective date · *ngày thay đổi có hiệu lực*. domain `vlo:LegalChange`; range `xsd:date`. The date from which a legal change takes effect, normally the effective date of the acting document.
- **`vlo:effectiveDate`** · effective date · *ngày có hiệu lực*. Sub-property of `eli:first_date_entry_in_force`; domain `vlo:LegalDocument`; range `xsd:date`. The date on which a legal document enters into force (source metadata first, then the commencement article).
- **`vlo:evidenceText`** · evidence text · *câu chứng cứ*. domain `vlo:LegalChange`; range `rdfs:Literal`. The operative sentence of the acting document from which the change was extracted, quoted in Vietnamese and cut at 300 characters.
- **`vlo:expiryDate`** · expiry date · *ngày hết hiệu lực*. Sub-property of `eli:date_no_longer_in_force`; domain `vlo:LegalDocument`; range `xsd:date`. The date from which a legal document is no longer in force, from the source status or from the effective date of the replacing or repealing act.
- **`vlo:isInForce`** · is in force · *còn hiệu lực*. range `xsd:boolean`. True if the document or term is known to be in force, false if known not to be in force; absent when unknown.
- **`vlo:mentionedDate`** · mentioned date · *ngày được nêu*. domain `vlo:LegalChange`; range `xsd:date`. The issue date of the affected document as stated in the evidence sentence, when given; used to disambiguate numbers without a year.
- **`vlo:ruleVersion`** · rule version · *phiên bản quy tắc trích xuất*. domain `vlo:LegalChange`; range `xsd:string`. The version of the extraction rules that produced a legal change, for example v7.
- **`vlo:sourceStatus`** · source status · *tình trạng theo nguồn*. range `rdfs:Literal`. The validity status string as published by the source, in Vietnamese (for example ‘Hết hiệu lực: 01/07/2025’).
- **`vlo:statusTextConflict`** · status-text conflict · *mâu thuẫn trạng thái và văn bản*. domain `vlo:LegalDocument`; range `xsd:boolean`. True when the source status says in force but an extracted relation says the document was replaced or repealed; such documents must be verified.

- **`vlo:validityEvidenceSource`** · validity evidence source · *nguồn căn cứ hiệu lực*. domain `vlo:LegalDocument`; range `xsd:string`. Which evidence determined the validity status: status, text, status+text or none.

Annotation properties

- **`vlo:shortcutProperty`** · shortcut property · *thuộc tính rút gọn*. Links a change kind to the direct document-to-document property that represents the same change.

Concept schemes

- **`vlo:ChangeKindScheme`** · Change kinds · *Loại thay đổi*. The five kinds of legal change extracted by the validity rules v7. Concepts: amended; partially terminated; partially repealed; repealed; replaced.
- **`vlo:ValidityStatusScheme`** · Validity statuses · *Tình trạng hiệu lực*. The eight validity values computed by the validity rules v7 as of the knowledge-graph date. Concepts: no longer in force; in force; in force, amended; no longer applicable; not yet in force; presumed in force, amended; superseded; unknown.

B.5 Module citation and legal basis (*trích dẫn và căn cứ*)

Legal-basis and citation links between legal resources and from other works to legal resources.

Object properties

- **`vlo:cites`** · cites · *trích dẫn*. Sub-property of `eli:cites`, `vlo:refersTo`; domain `eli:LegalResource`; range `eli:LegalResource`. A legal document, or one of its articles, cites another legal document or a provision of it. Self-references and anaphoric references are not included.
- **`vlo:citesParentLaw`** · cites parent law · *trích dẫn luật gốc*. Sub-property of `vlo:cites`; domain `eli:LegalResource`; range `eli:LegalResource`. A citation that the source dataset marks as a reference to the parent law of the citing document, that is, the law the citing document details or implements.
- **`vlo:isLegalBasisOf`** · is legal basis of · *là căn cứ của*. domain `vlo:LegalDocument`; range `vlo:LegalDocument`. Inverse of vlo:legalBasis. Not materialised in the RDF dumps.
- **`vlo:legalBasis`** · legal basis · *căn cứ pháp lý*. Sub-property of `eli:based_on`, `vlo:refersTo`; domain `vlo:LegalDocument`; range `vlo:LegalDocument`; inverse of `vlo:isLegalBasisOf`. The object document is a legal basis on which the subject document was issued, as listed in the 'Căn cứ …' recitals of its preamble or in the source's parent-acts field. The object never has a lower legal-force rank than the subject.
- **`vlo:refersTo`** · refers to · *dẫn chiếu*. Sub-property of `dcterms:references`. The most general VLOnto reference link: a work (legal document, provision, court decision, answer or news article) refers to a legal resource or judicial document. Specialised by the citation, basis, application and mention properties.

B.6 Module judiciary (*tư pháp*)

Court decisions, judgments, precedents and their classification. *The terms of this module are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no court data.* (*Các khái niệm của mô-đun này có dữ liệu từ VLOnto 2.0 (các bộ dữ liệu tòa án anle-toaan-gov-vn và cbba-toaan-gov-vn); đồ thị tri thức VLOnto 1.0.0 chưa có dữ liệu tòa án.*)

Classes

- **`vlo:CaseCategory`** · Case category · *Loại vụ việc*. Subclass of `skos:Concept`. The legal domain of a case: civil, criminal, administrative, commercial, labour, marriage and family, bankruptcy or economic. *Scope: Its concepts are defined in the ontology; they classify court decisions, which are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn). VLOnto 1.0.0 has no court decisions.*
- **`vlo:CourtDecision`** · Court decision · *Bản án, quyết định của Tòa án*. Subclass of `vlo:JudicialDocument`; disjoint with `vlo:Precedent`. A judgment or ruling issued by a court in a specific case, identified by its decision number (for example 38/2021/DS-PT), court and date. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*

- **`vlo:CourtLevel`** · Court level · *Cấp Tòa án*. Subclass of `skos:Concept`. The level of a court in the court system: supreme, high, provincial, district or military. *Scope: Its concepts are defined in the ontology; they classify court decisions, which are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn). VLOnto 1.0.0 has no court decisions.*
- **`vlo:CourtRuling`** · Court ruling · *Quyết định của Tòa án*. Subclass of `vlo:CourtDecision`. A decision of a court other than a judgment, for example recognising the parties' agreement or suspending a case. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:InstanceLevel`** · Instance level · *Cấp xét xử*. Subclass of `skos:Concept`. The procedural stage at which a decision is made: first instance, appeal, cassation review or retrial review. *Scope: Its concepts are defined in the ontology; they classify court decisions, which are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn). VLOnto 1.0.0 has no court decisions.*
- **`vlo:Judgment`** · Judgment · *Bản án*. Subclass of `vlo:CourtDecision`; disjoint with `vlo:CourtRuling`. A court decision that settles a case on its merits after a hearing (first instance, appeal, cassation or retrial). *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:JudicialDocument`** · Judicial document · *Văn bản của Tòa án*. Subclass of `schema:CreativeWork`; disjoint with `vlo:LegalDocument`. A document produced in the judicial system: a court decision in a case or a published case-law precedent. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:Precedent`** · Precedent · *Án lệ*. Subclass of `vlo:JudicialDocument`. Reasoning or a ruling in a court decision that the Council of Judges of the Supreme People's Court has selected and the Chief Justice has published as a precedent for courts to follow; identified by its number, for example 04/2016/AL. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*

Object properties

- **`vlo:appliesDocument`** · applies document · *áp dụng văn bản*. Sub-property of `vlo:refersTo`; domain `vlo:JudicialDocument`; range `vlo:LegalDocument`. A court decision cites a legal document as a whole, or a provision of it that is not available as an article node. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:appliesPrecedent`** · applies precedent · *áp dụng án lệ*. Sub-property of `vlo:citesCase`; domain `vlo:JudicialDocument`; range `vlo:Precedent`. A court decision cites a precedent by its number. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:appliesProvision`** · applies provision · *áp dụng điều khoản*. Sub-property of `vlo:refersTo`; domain `vlo:JudicialDocument`; range `vlo:Provision`. A court decision cites an article, clause or point of a legal document as applied or relied on. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:caseCategory`** · case category · *loại vụ việc*. domain `vlo:JudicialDocument`; range `vlo:CaseCategory`. The legal domain of the case decided or of the precedent. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:citesCase`** · cites case · *viện dẫn bản án*. Sub-property of `vlo:refersTo`; domain `vlo:JudicialDocument`; range `vlo:JudicialDocument`. A court decision refers to another court decision (for example the first-instance judgment under appeal) or to a precedent. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:courtLevel`** · court level · *cấp Tòa án*. range `vlo:CourtLevel`. The level of the court that issued a decision, or of a court. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:decidedBy`** · decided by · *Tòa án ban hành*. Sub-property of `dcterms:creator`; domain `vlo:CourtDecision`; range `vlo:Court`. The court that issued a court decision. *Scope: Populated from*

VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.

- **`vlo:instanceLevel`** · instance level · *cấp xét xử*. domain `vlo:CourtDecision`; range `vlo:InstanceLevel`. The procedural stage at which a court decision was made. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:selectedFrom`** · selected from · *được lựa chọn từ*. Sub-property of `prov:wasDerivedFrom`; domain `vlo:Precedent`; range `vlo:CourtDecision`. The court decision whose reasoning or ruling was selected as the precedent. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*

Datatype properties

- **`vlo:decisionCode`** · decision code · *ký hiệu bản án, quyết định*. domain `vlo:CourtDecision`; range `xsd:string`. The symbol part of a decision number that encodes case domain and stage, for example DS-PT (civil, appeal) or QĐST-HNGĐ (first-instance ruling, marriage and family). *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:decisionDate`** · decision date · *ngày tuyên án, ra quyết định*. Sub-property of `dcterms:date`; domain `vlo:CourtDecision`; range `xsd:date`. The date on which a court decision was issued. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:decisionNumber`** · decision number · *số bản án, quyết định*. Sub-property of `dcterms:identifier`; domain `vlo:CourtDecision`; range `xsd:string`. The number of a court decision as issued, for example 38/2021/DS-PT. Numbers are unique only within one court. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*
- **`vlo:precedentNumber`** · precedent number · *số án lệ*. Sub-property of `dcterms:identifier`; domain `vlo:Precedent`; range `xsd:string`. The national number of a precedent, for example 04/2016/AL. *Scope: Populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn); the VLOnto 1.0.0 knowledge graph has no data for this term.*

Concept schemes

- **`vlo:CaseCategoryScheme`** · Case categories · *Loại vụ việc*. Legal domains of cases as recorded by the court-decision datasets. Concepts: administrative; bankruptcy; civil; business and commercial; criminal; economic; labour; marriage and family. *Scope: Its concepts are defined in the ontology; they classify court decisions, which are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn). VLOnto 1.0.0 has no court decisions.*
- **`vlo:CourtLevelScheme`** · Court levels · *Cấp Tòa án*. Levels of the Vietnamese court system as recorded by the court-decision datasets. Concepts: district court; High People's Court; military court; provincial court; Supreme People's Court. *Scope: Its concepts are defined in the ontology; they classify court decisions, which are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn). VLOnto 1.0.0 has no court decisions.*
- **`vlo:InstanceLevelScheme`** · Instance levels · *Cấp xét xử*. Procedural stages at which court decisions are made. Concepts: appeal; cassation review; first instance; retrial review. *Scope: Its concepts are defined in the ontology; they classify court decisions, which are populated from VLOnto 2.0 (court datasets anle-toaan-gov-vn and cbba-toaan-gov-vn). VLOnto 1.0.0 has no court decisions.*

B.7 Module legal Q&A (*hỏi đáp pháp luật*)

Published legal questions and their answers.

Classes

- **`vlo:LegalAnswer`** · Legal answer · *Câu trả lời pháp luật*. Subclass of `schema:Answer`. The published answer to a legal question; in the knowledge graph it carries the citations of legal documents and provisions found in the answer, not its text.

- **`vlo:LegalQuestion`** · Legal question · *Câu hỏi pháp luật*. Subclass of `schema:Question`; disjoint with `vlo:LegalAnswer`. A published legal question from a legal Q&A service. The knowledge graph keeps its identifier, link, date and legal area, not its text.

Object properties

- **`vlo:answerCitesDocument`** · answer cites document · *câu trả lời trích dẫn văn bản*. Sub-property of `schema:citation`, `vlo:refersTo`; domain `vlo:LegalAnswer`; range `vlo:LegalDocument`. A legal answer cites a legal document by its number.
- **`vlo:answerCitesProvision`** · answer cites provision · *câu trả lời trích dẫn điều khoản*. Sub-property of `schema:citation`, `vlo:refersTo`; domain `vlo:LegalAnswer`; range `vlo:Provision`. A legal answer cites an article, clause or point of a legal document.
- **`vlo:hasAnswer`** · has answer · *có câu trả lời*. Sub-property of `schema:acceptedAnswer`; domain `vlo:LegalQuestion`; range `vlo:LegalAnswer`; `owl:FunctionalProperty`. Links a legal question to its published answer.

Datatype properties

- **`vlo:publishedDate`** · published date · *ngày đăng*. Sub-property of `dcterms:issued`; range `xsd:date`. The date on which a question, answer or news article was published by its source.

B.8 Module terminology (*thuật ngữ*)

Legal terms organised as SKOS concepts.

Classes

- **`vlo:LegalTerm`** · Legal term · *Thuật ngữ pháp lý*. Subclass of `skos:Concept`. A term of Vietnamese legal language, organised as a SKOS concept with a Vietnamese preferred label, an English label (machine translation unless marked otherwise), a legal area, a status and related terms.

Object properties

- **`vlo:relatedTerm`** · related term · *thuật ngữ liên quan*. Sub-property of `skos:related`; domain `vlo:LegalTerm`; range `vlo:LegalTerm`; `owl:SymmetricProperty`. A legal term listed by the source as related to another term.

Datatype properties

- **`vlo:englishLabelSource`** · English label source · *nguồn nhãn tiếng Anh*. domain `vlo:LegalTerm`; range `xsd:string`. Provenance of the English label of a term: ‘mt’ for machine translation, ‘site’ for an English name given by the source site.
- **`vlo:translationModel`** · translation model · *mô hình dịch*. domain `vlo:LegalTerm`; range `xsd:string`. Identifier of the model that machine-translated the English label, as recorded by the source dataset.

Concept schemes

- **`vlo:LegalTermScheme`** · Vietnamese legal terminology · *Thuật ngữ pháp lý Việt Nam*. The glossary of legal terms derived from the terminology dataset; populated by the knowledge graph.

B.9 Module news (*tin tức*)

Legal news and policy articles that mention legal documents.

Classes

- **`vlo:LegalNewsArticle`** · Legal news article · *Bài viết tin tức pháp luật*. Subclass of `schema:NewsArticle`. A news, policy or advice article about law published by a legal information portal; the knowledge graph keeps its identifier, link, date, category and the legal documents it mentions.
- **`vlo:NewsCategory`** · News category · *Chuyên mục tin tức*. Subclass of `skos:Concept`. A section of a legal news portal, such as new policies or real estate.

Object properties

- **`vlo:mentionsDocument`** · mentions document · *nhắc đến văn bản*. Sub-property of `schema:mentions`, `vlo:refersTo`; domain `vlo:LegalNewsArticle`; range `vlo:LegalDocument`. A news article mentions a legal document by its number.

- **`vlo:newsCategory`** · news category · *chuyên mục*. domain `vlo:LegalNewsArticle`; range `vlo:NewsCategory`. The section of the portal in which a news article was published.

Concept schemes

- **`vlo:NewsCategoryScheme`** · News categories · *Chuyên mục tin tức*. Sections of the legal news portal; populated by the knowledge graph.

B.10 Module classification (*phân loại*)

Concept schemes for legal areas, document types and legal-force ranks.

Classes

- **`vlo:DocumentType`** · Document type · *Loại văn bản*. Subclass of `eli:ResourceType`, `skos:Concept`. A type of legal document, such as law, decree or circular; used as the value of eli:type_document.
- **`vlo:LegalArea`** · Legal area · *Lĩnh vực pháp luật*. Subclass of `skos:Concept`. A subject area of law used to classify documents, questions and terms, such as labour and wages or real estate.
- **`vlo:LegalForceRank`** · Legal-force rank · *Thứ bậc hiệu lực pháp lý*. Subclass of `skos:Concept`. A level in the hierarchy of legal force of Vietnamese normative documents, from the Constitution (level 0, highest) to circulars and ministerial decisions (level 5, lowest). An act cannot replace, repeal or amend an act of a higher level.

Object properties

- **`vlo:legalArea`** · legal area · *lĩnh vực*. Sub-property of `dcterms:subject`; range `vlo:LegalArea`. The legal area under which a document, question or term is classified by its source.
- **`vlo:legalForceRank`** · legal-force rank · *thứ bậc hiệu lực*. domain `vlo:LegalDocument`; range `vlo:LegalForceRank`; `owl:FunctionalProperty`. The level of a legal document in the hierarchy of legal force, derived from its type and issuer (rule 5 of the validity rules v7).

Datatype properties

- **`vlo:rankLevel`** · rank level · *bậc*. domain `vlo:LegalForceRank`; range `xsd:integer`. The numeric level of a legal-force rank: 0 is the highest (Constitution), 5 the lowest.

Annotation properties

- **`vlo:typeConcept`** · type concept · *khái niệm loại văn bản*. Links a VLOnto document class to the concept of the DocumentType scheme that represents the same document type, so that class membership and the eli:type_document value can be kept in step.

Concept schemes

- **`vlo:DocumentTypeScheme`** · Document types · *Loại văn bản*. The ten types of normative legal documents in the scope of VLOnto. Concepts: Circular; Code; Constitution; Decision; Decree; Joint circular; Joint resolution; Law; Ordinance; Resolution.
- **`vlo:LegalAreaScheme`** · Legal areas (documents and Q&A) · *Lĩnh vực pháp luật (văn bản và hỏi đáp)*. The 27 legal areas used by the document and Q&A datasets; populated by the knowledge graph.
- **`vlo:LegalForceRankScheme`** · Legal-force ranks · *Thứ bậc hiệu lực pháp lý*. Six levels of legal force used by the validity rules (rule 5) and as the vertical layer of the 3D legal network. Concepts: Constitution; Codes, laws and National Assembly resolutions; Ordinances and Standing Committee resolutions; Decrees, joint resolutions and other resolutions; Prime Minister decisions; Circulars, joint circulars and other decisions.
- **`vlo:TermAreaScheme`** · Legal areas (terminology) · *Lĩnh vực pháp luật (thuật ngữ)*. The legal areas used by the terminology dataset; areas with the same name as a document area are the same concept. Populated by the knowledge graph.

B.11 Module provenance (*nguồn gốc*)

Source datasets, extraction methods, edge sets and build activities.

Classes

- **`vlo:BuildActivity`** · Build activity · *Lần xây dựng đồ thị*. Subclass of `prov:Activity`. One run of the VLOnto build pipeline that generated the knowledge-graph tables and RDF dumps.

- **`vlo:EdgeSet`** · Edge set · *Tập liên kết*. Subclass of `prov:Entity`. The set of all knowledge-graph links of one property produced from one source, described with its size, the extraction methods used and the build that generated it. Per-link methods are kept in the parquet edge tables.
- **`vlo:ExtractionMethod`** · Extraction method · *Phương pháp trích xuất*. Subclass of `prov:Plan`, `skos:Concept`. A documented procedure that produced records or links of the knowledge graph, for example a rule set or a resolver strategy.
- **`vlo:Module`** · Module · *Mô-đun bản thể*. Subclass of `skos:Concept`. A thematic group of VLOnto terms, such as documents, provisions or judiciary.
- **`vlo:SourceDataset`** · Source dataset · *Bộ dữ liệu nguồn*. Subclass of `prov:Entity`. A published dataset, pinned to a revision, from which knowledge-graph records were derived; carries its licence and attribution text.

Object properties

- **`vlo:edgeProperty`** · edge property · *thuộc tính liên kết*. domain `vlo:EdgeSet`. The property used by all links of an edge set.
- **`vlo:extractionMethod`** · extraction method · *phương pháp trích xuất*. range `vlo:ExtractionMethod`. The extraction method that produced a legal change or the links of an edge set.
- **`vlo:sourceDataset`** · source dataset · *bộ dữ liệu nguồn*. Sub-property of `dcterms:source`; range `vlo:SourceDataset`. The source dataset from which a knowledge-graph record was derived.

Datatype properties

- **`vlo:attributionText`** · attribution text · *lời ghi nguồn*. domain `vlo:SourceDataset`; range `rdfs:Literal`. The attribution that must accompany material derived from the dataset.
- **`vlo:datasetRevision`** · dataset revision · *phiên bản bộ dữ liệu*. domain `vlo:SourceDataset`; range `xsd:string`. The commit hash of the dataset revision that was used.
- **`vlo:edgeCount`** · edge count · *số liên kết*. domain `vlo:EdgeSet`; range `xsd:integer`. The number of links in an edge set.
- **`vlo:inclusionPolicy`** · inclusion policy · *chính sách sử dụng*. domain `vlo:SourceDataset`; range `rdfs:Literal`. What VLOnto takes from the dataset and what it leaves out, following the dataset licence.
- **`vlo:licenceLabel`** · licence label · *nhãn giấy phép*. domain `vlo:SourceDataset`; range `xsd:string`. The licence identifier given in the card of the source dataset, for example cc-by-4.0 or other.
- **`vlo:softwareVersion`** · software version · *phiên bản phần mềm*. domain `vlo:BuildActivity`; range `xsd:string`. The version of the VLOnto build code used by a build activity.
- **`vlo:sourceId`** · source identifier · *mã trong nguồn*. Sub-property of `dcterms:identifier`; range `xsd:string`. The identifier of the record in its source dataset (for example the vbpl document id 333670 or the cbba doc_name).
- **`vlo:sourceUrl`** · source URL · *đường dẫn nguồn*. Sub-property of `schema:url`; range `xsd:anyURI`. The web address of the record at its original publisher, where its full text can be read.
- **`vlo:validityAsOf`** · validity as of · *hiệu lực tính đến ngày*. domain `vlo:BuildActivity`; range `xsd:date`. The reference date at which document validity was computed.

Annotation properties

- **`vlo:module`** · module · *mô-đun*. range `vlo:Module`. The thematic module of VLOnto to which a class, property or concept scheme belongs; used to group and colour terms in documentation and in the 3D ontology playground.

Concept schemes

- **`vlo:ExtractionMethodScheme`** · Extraction methods · *Phương pháp trích xuất*. Procedures that produced VLOnto records and links; the notation is the value used in the method columns of the parquet edge tables. Concepts: answer text, name; answer text, number; case number and court; case number and date; article parser; citation field, name; Constitution in the preamble; court name; document-number key; document-number pattern; issuer code; law name and year; parent-acts field; preamble citation; precedent number; source metadata; unique document of type; validity rules v7.

- **`vlo:ModuleScheme`** · VLOnto modules · *Các mô-đun của VLOnto*. The eleven thematic modules of VLOnto. Concepts: agents; citation and legal basis; classification; documents; judiciary; news; provenance; provisions; legal Q&A; terminology; validity and change.

Appendix C. Data dictionary

Generated from `kg/data_dictionary.json`. “Non-null” counts the rows with a value. The table marked local-only is written by the build for local use only and is never shared (Appendix D).

C.1 `documents` (40,816 rows)

Legal documents (40,816 in-scope normative documents from vbpl, after cleaning).

Column	Type	Non-null	Description
`doc_id`	String	40,816	vbpl document id (primary key)
`iri`	String	40,816	VLOnto IRI
`doc_type_vi`	String	40,816	Vietnamese document type
`doc_class`	String	40,816	VLOnto class local name (Law, Decree, ...)
`doc_type_code`	String	40,816	type notation (L, NĐ, TT, ...)
`doc_number`	String	40,215	official number, e.g. 45/2019/QH14 (null for 601 documents)
`doc_number_key`	String	40,215	normalised lookup key
`title_vi`	String	40,816	Vietnamese title
`title_core`	String	36,984	title without the type word (used for name resolution)
`issue_date`	String	40,502	issue date (ISO); null when it is more than one year away from the year in the number
`issue_date_check`	String	40,816	ok, missing or year_mismatch_with_number
`issue_date_source`	String	144	source issue date when it was rejected
`year`	Int64	40,720	issue year (from the number when the date was rejected)
`effective_date`	String	22,076	entry into force (metadata, then text)
`effective_date_src`	String	22,076	source of effective_date
`expiry_date`	String	7,256	no longer in force from
`status_raw`	String	4,356	source status string (Vietnamese)
`status_code`	String	4,356	normalised source status
`status_date`	String	2,207	date in the status string
`validity`	String	40,816	validity v7 (8 values)
`in_force`	Boolean	12,156	true/false/null
`status_text_conflict`	Boolean	40,816	source says in force but text says superseded
`validity_evidence`	String	40,816	status / text / status+text / none
`legal_force_rank`	Int64	40,816	0 (Constitution) .. 5 (circulars)
`area_id`	String	40,816	legal area slug
`issuer_ids`	List(String)	40,816	canonical issuer codes (joint documents have several)
`issuer_method`	String	40,816	issuer-code (symbol of the number), doc-type-rule (laws, codes, constitutions -> QH; ordinances), source-metadata, code-unknown, none
`issuer_codes_unmatched`	List(String)	40,816	symbol parts of the number that are not a known issuer (kept for transparency, no node)
`tier`	Int64	40,816	vbpl tier (0 primary legislation, 1 central decisions)
`url`	String	40,816	source URL (thuvienphapluat.vn)
`n_chapters`	Int64	40,816	chapters found by the parser
`has_articles`	Boolean	40,816	article structure parsed
`source_dataset`	String	40,816	source dataset
`n_articles_main`	UInt32	40,816	main-body articles
`n_articles_appendix`	UInt32	40,816	appendix articles

C.2 `articles` (436,309 rows)

Articles (Điều): one node per main-body or appendix article; parts of split articles joined. No text (licence).

Column	Type	Non-null	Description
`article_id`	String	436,309	<doc_id>::dieu-<n> or <doc_id>::att<s>-dieu-<n> (chunk unit id)
`iri`	String	436,309	VLOnto IRI
`doc_id`	String	436,309	document
`article_no`	String	436,309	article number as printed (5, 5a)
`segment`	String	436,309	main or att<s> (appendix s)
`heading`	String	349,840	article heading (<=200 chars)
`parent_id`	String	436,309	immediate parent (structure id or doc_id)
`parent_kind`	String	436,309	Document/Appendix/Part/Chapter/Section
`position`	Int64	436,309	order of the article in its document
`n_chunks`	Int64	436,309	retrieval chunks of the article
`chars`	Int64	436,309	characters of text
`chapter_heading`	String	315,885	chapter heading
`section_heading`	String	82,344	section heading

C.3 `structure` (73,925 rows)

Appendices, parts, chapters and sections of documents.

Column	Type	Non-null	Description
`struct_id`	String	73,925	<doc_id>::<path> e.g. 333670::chuong-iii/muc-3
`doc_id`	String	73,925	document
`kind`	String	73,925	Appendix/Part/Chapter/Section
`number`	String	73,925	number token (iii, 3) or o<k> when not parsable
`heading`	String	73,234	heading (<=200 chars)
`parent_id`	String	73,925	parent node
`parent_kind`	String	73,925	kind of parent
`position`	Int64	73,925	order among nodes of this kind in the document
`iri`	String	73,925	VLOnto IRI

C.4 `provisions_ref` (48,031 rows)

Clauses (khoản) and points (điểm) created because an edge refers to them.

Column	Type	Non-null	Description
`prov_id`	String	48,031	<article_id>::khoan-<k>[::diem-<p>] or <article_id>::diem-<p>
`iri`	String	48,031	VLOnto IRI
`kind`	String	48,031	Clause or Point
`article_id`	String	48,031	article
`doc_id`	String	48,031	document
`clause`	String	47,957	clause number
`point`	String	12,555	point letter
`parent_id`	String	48,031	article or clause
`n_refs`	Int64	48,031	edges that reference this provision

C.5 `issuers` (57 rows)

Issuing bodies identified by the symbol of the document number (curated table; unknown symbols get no node).

Column	Type	Non-null	Description
`issuer_id`	String	57	canonical ASCII code (BTC, TTG, QH, ...)
`iri`	String	57	VLOnto IRI
`label_vi`	String	57	Vietnamese name
`label_en`	String	57	English name
`kind`	String	57	ministry, government, national-assembly, ...
`label_source`	String	57	vlonto-table
`code_aliases`	List(String)	57	other symbols mapped to this body
`n_documents`	Int64	57	documents issued

C.6 `areas` (51 rows)

Legal areas (SKOS concepts) from vbpl/hdpl (27) and tnpl.

Column	Type	Non-null	Description
`area_id`	String	51	slug
`iri`	String	51	VLOnto IRI
`label_vi`	String	51	Vietnamese label
`label_en`	String	51	English label
`label_en_source`	String	51	vlonto (written for VLOnto) or tnpl-mt
`schemes`	List(String)	51	LegalAreaScheme and/or TermAreaScheme
`n_documents`	Int64	51	documents
`n_qas`	Int64	51	Q&As
`n_terms`	Int64	51	terms

C.7 `doc_types` (10 rows)

Document types (concepts of the ontology).

Column	Type	Non-null	Description
`doc_type_vi`	String	10	-
`label_en`	String	10	-
`class`	String	10	-
`concept`	String	10	-
`notation`	String	10	-
`n_documents`	Int64	10	-

C.8 `legal_force_ranks` (6 rows)

Legal-force ranks (concepts of the ontology).

Column	Type	Non-null	Description
`rank`	Int64	6	-
`concept`	String	6	-
`label_en`	String	6	-
`label_vi`	String	6	-
`n_documents`	Int64	6	-

C.9 changes (15,035 rows)

Legal change events (validity relations v7), one per relation.

Column	Type	Non-null	Description
change_id	String	15,035	<acting>-<kind>-<affected>
iri	String	15,035	VLOnto IRI
acting_doc_id	String	15,035	acting (newer) document
affected_doc_id	String	15,035	affected document
kind	String	15,035	replaced/amended/repealed/partially_repealed/ partially_expired
evidence	String	15,035	operative sentence (<=300 chars)
effective_date	String	15,002	effective date of the change
mentioned_date	String	12,948	date of the affected act in the sentence
acting_number	String	14,913	number of acting act
affected_number	String	15,035	number of affected act
affected_number_as_cited	String	15,035	number as written
method	String	15,035	validity-rules-v7
rule_version	String	15,035	v7

C.10 edges_change (15,035 rows)

Shortcut edges acting document -> affected document (one per change event).

Column	Type	Non-null	Description
src_doc_id	String	15,035	acting document
dst_doc_id	String	15,035	affected document
kind	String	15,035	change kind
property	String	15,035	vlo property
change_id	String	15,035	event
method	String	15,035	validity-rules-v7

C.11 edges_legal_basis (99,038 rows)

Legal basis: citing document -> basis document.

Column	Type	Non-null	Description
src_doc_id	String	99,038	citing document
dst_doc_id	String	99,038	basis document
method	String	99,038	parent-acts, preamble-citation or constitution-preamble (first in this order when several apply)
methods	List(String)	99,038	all methods that found the edge
resolve_method	String	99,038	docnum-key or law-name-year
raw	String	99,038	reference as written

C.12 `edges_citation` (268,563 rows)

vbpl citations (external, external_mention, parent_law, parent_law_mention) resolved to documents/articles.

Column	Type	Non-null	Description
src_doc_id	String	268,563	citing document
src_article_id	String	133,431	citing article when known
dst_doc_id	String	268,563	cited document
dst_article_id	String	97,153	cited article when given and present
clause	String	56,649	cited clause
point	String	17,466	cited point
kind	String	268,563	vbpl citation kind
n_mentions	UInt32	268,563	records merged into this edge
article_cited	String	98,120	article number written (even if absent)
source_location	String	268,563	where in the citing document
raw	String	268,563	citation as written (<=200 chars)
method	String	268,563	docnum-key or law-name-year

C.13 `qas` (109,543 rows)

Legal Q&As from hdpl: metadata only (no question or answer text).

Column	Type	Non-null	Description
qa_id	String	109,543	hdpl id
iri	String	109,543	question IRI
answer_iri	String	109,543	answer IRI
url	String	109,543	source URL
published_date	String	109,543	publication date
year	Int64	109,543	publication year
category	String	109,482	category slug
area_id	String	109,482	legal area
has_ad	Boolean	109,543	advertising flag of the source
n_citation_records	Int64	109,543	citation records
n_document_citations	Int64	109,543	document-level records
n_provision_citations	Int64	109,543	provision-level records
n_gold_articles	UInt32	109,543	articles linked (gold resolver)
n_documents_resolved	UInt32	109,543	documents linked by number

C.14 `edges_qa_article` (34,869 rows)

Q&A answer -> article (+clause, point) from the gold resolver (gold_all.parquet).

Column	Type	Non-null	Description
qa_id	String	34,869	Q&A
dst_doc_id	String	34,869	document
dst_article_id	String	34,869	article
clause	String	16,424	clause
point	String	3,221	point
ref	String	34,869	reference as written
method	String	34,869	answer-text-number / answer-text-name / citation-field-name / unique-document-of-type
method_detail	String	34,869	gold method string

C.15 `edges_qa_document` (119,683 rows)

Q&A answer -> document cited by number.

Column	Type	Non-null	Description
`qa_id`	String	119,683	Q&A
`dst_doc_id`	String	119,683	document
`number_cited`	String	119,683	number as written
`law_type`	String	119,683	law type
`method`	String	119,683	docnum-key

C.16 `terms` (16,701 rows)

Legal terms from tnpl (names only, no definitions).

Column	Type	Non-null	Description
`term_id`	Int64	16,701	tnpl id
`iri`	String	16,701	VLOnto IRI
`label_vi`	String	16,701	Vietnamese term
`label_en`	String	16,599	English label
`label_en_source`	String	16,599	mt or site
`translation_model`	String	15,426	MT model
`area_id`	String	16,700	area slug
`tnpl_area_id`	Int64	16,700	tnpl area id
`status_vi`	String	16,701	status
`in_force`	Boolean	16,701	status as boolean
`url`	String	16,701	source URL

C.17 `edges_term_related` (17,615 rows)

Related terms.

Column	Type	Non-null	Description
`src_term_id`	Int64	17,615	term
`dst_term_id`	Int64	17,615	related term
`method`	String	17,615	source-metadata

C.18 `news` (681 rows)

Legal news articles from htpl (no title or text).

Column	Type	Non-null	Description
`news_id`	String	681	htpl id
`iri`	String	681	VLOnto IRI
`url`	String	681	source URL
`category`	String	681	category slug
`published_date`	String	87	date
`year`	Int64	87	year
`n_numbers_found`	Int64	681	document numbers found
`n_documents_resolved`	Int64	681	documents linked

C.19 `edges_news_document` (1,484 rows)

News article -> mentioned document.

Column	Type	Non-null	Description
`news_id`	String	1,484	news
`dst_doc_id`	String	1,484	document
`number_found`	String	1,484	number
`method`	String	1,484	docnum-regex

C.20 `news_categories` (7 rows)

News categories.

Column	Type	Non-null	Description
`category`	String	7	-
`iri`	String	7	-
`label_vi`	String	7	-
`label_en`	String	7	-
`n_articles`	Int64	7	-

C.21 `sources` (4 rows)

Source datasets of this build with revision, licence and policy (release v1: the four thuvienphapluat.vn datasets).

Column	Type	Non-null	Description
`dataset`	String	4	-
`iri`	String	4	-
`revision`	String	4	-
`licence`	String	4	-
`publisher`	String	4	-
`content`	String	4	-
`repository`	String	4	-
`card_sha256`	String	4	-

C.22 `article_text` (436,309 rows, local-only)

Column	Type	Non-null	Description
`article_id`	String	436,309	-
`text`	String	436,309	-

Appendix D. Licences and attribution

Sources. Licence policy version 1.0.0 (release v1). Decided 2026-11-11 for 0.1.0 from the cards (`README.md`) of the source datasets, downloaded at the pinned revisions below; revised 2026-11-12 for release v1, which is built from the four thuvienphapluat.vn datasets only. The two court datasets (`anle-toaan-gov-vn`, `cbba-toaan-gov-vn`) are planned for v2: v1 reads nothing from them, and their rules (policy section 2.5) apply again when they are added (build switch `VLONTO_INCLUDE_COURTS=1`). The policy is `docs/licence_policy.md`; it is a reading of the dataset cards by the build, not legal advice.

Dataset	Card licence	Original source
`tmquan/thuvienphapluat-vn-hdpl`	other	thuvienphapluat.vn
`tmquan/thuvienphapluat-vn-htpl`	other	thuvienphapluat.vn
`tmquan/thuvienphapluat-vn-tnpl`	other	thuvienphapluat.vn/tnpl
`tmquan/thuvienphapluat-vn-vbpl`	cc-by-4.0	thuvienphapluat.vn

What the outputs contain. The tables, the RDF dumps and the website data hold metadata, titles, article and chapter headings (up to 200 characters), evidence sentences of change events (up to 300 characters), identifiers, URLs, dates, categories, term names and the edges derived by VLOnto. They hold no article or document bodies, Q&A or news text, glossary definitions or person names. The one exception is the website's question traces page, which shows the published titles of the questions it traces (section 9). The article text is kept in a local-only table (`article_text`, 436,309 rows) that the build adds to the articles table (`--with-article-text`) only if the owner decides that sharing it is acceptable.

Licence of the VLOnto outputs.

- Ontology (`vlonto.ttl`, `vlonto-shapes.ttl`, documentation): **CC BY 4.0**, creator tmquan.
- Website Traces page data (`site/data/traces/`): VLOnto's traces, scores and drawings under **CC BY 4.0**; the titles of the 538 traced questions it holds remain subject to the source terms of thuvienphapluat.vn (see policy section 2.2).
- Knowledge-graph tables and RDF dumps: VLOnto's own contributions (resolution, structure, derived edges, validity) are released under **CC BY 4.0**; records derived from the `other`-licensed sources (`hdpl`, `tnpl`, `htpl`) contain only identifiers, links, dates, categories, short labels and derived edges, and remain subject to the source terms of thuvienphapluat.vn.
- Every source dataset of the build is described in the graph as a `vlo:SourceDataset` with revision, licence IRI and attribution text, and every node carries `vlo:sourceDataset`. Release v1 describes the four thuvienphapluat.vn datasets; the court datasets appear from v2.

Attribution (use on the website, in the repository and in derived work):

> VLOnto is derived from four datasets published on Hugging Face by TMQuan: `thuvienphapluat-vn-vbpl` (metadata CC BY 4.0; source thuvienphapluat.vn), `thuvienphapluat-vn-hdpl`, `thuvienphapluat-vn-tnpl` and `thuvienphapluat-vn-htpl` (source thuvienphapluat.vn and thuvienphapluat.vn/tnpl/; only identifiers, links, dates, categories, term names and derived links are used). English term labels from `tnpl` are machine translations. Extracted relations are produced automatically and may be wrong. VLOnto is not an official legal text and is not legal advice.

> VLOnto được xây dựng từ bốn bộ dữ liệu do TMQuan công bố trên Hugging Face: `thuvienphapluat-vn-vbpl` (siêu dữ liệu theo CC BY 4.0; nguồn thuvienphapluat.vn), `thuvienphapluat-vn-hdpl`, `thuvienphapluat-vn-tnpl` và `thuvienphapluat-vn-htpl` (nguồn thuvienphapluat.vn; chỉ dùng mã định danh, liên kết, ngày, lĩnh vực, tên thuật ngữ và các liên kết suy ra). Tên tiếng Anh của thuật ngữ là bản dịch máy. Các quan hệ được trích xuất tự động và có thể sai. VLOnto không phải văn bản pháp luật chính thức và không phải tư vấn pháp lý.

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